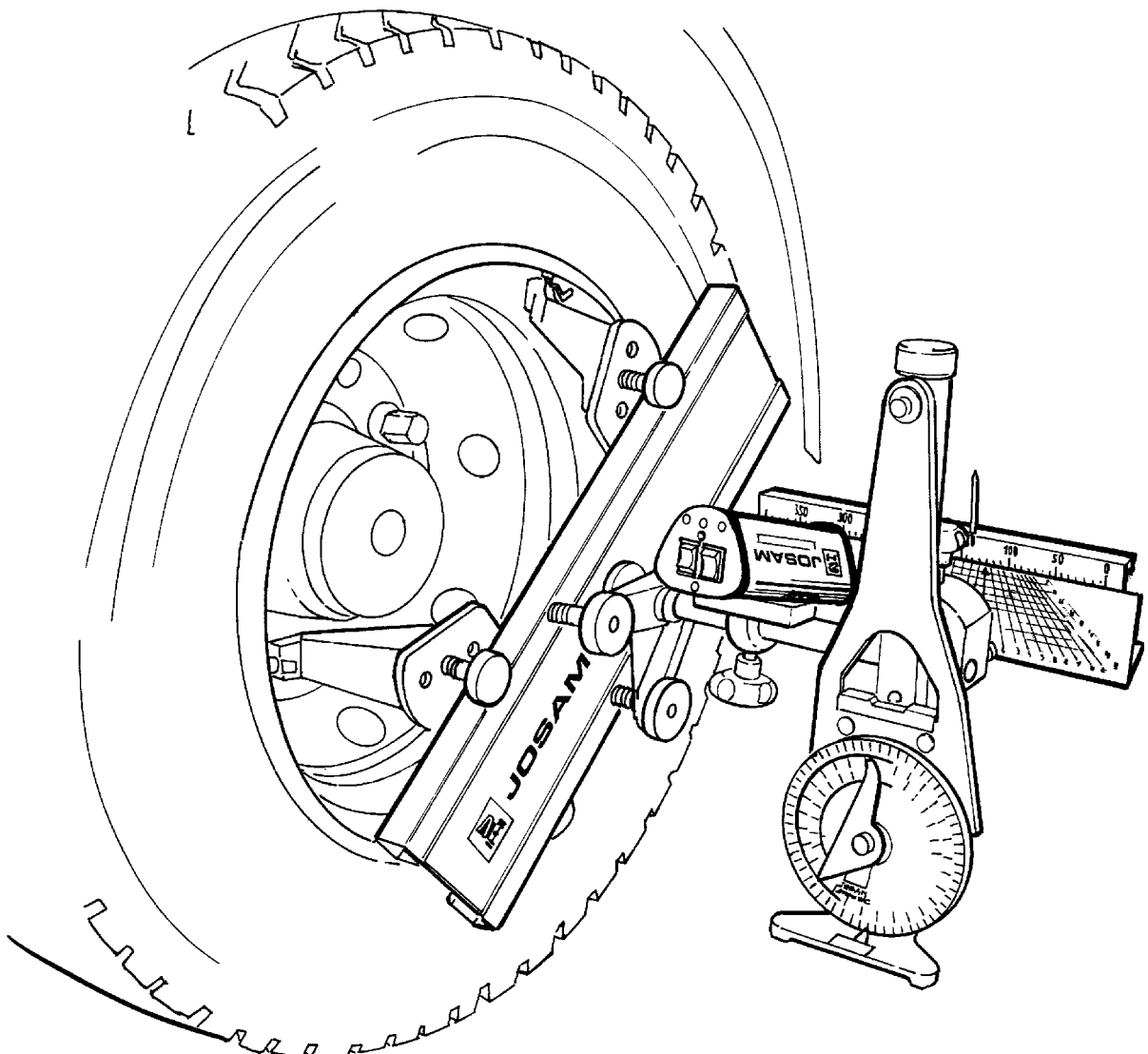




JOSAM[®] laser AM Measuring system

ALIGNMENT OF WHEELS AND AXLES

Manual[©]

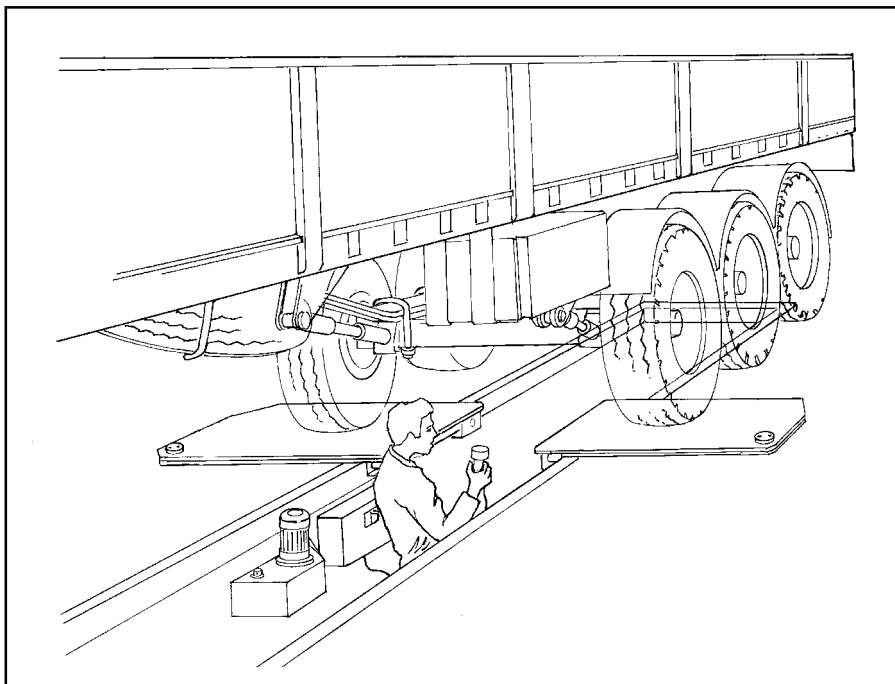




The following things should be checked before wheel and axle measurements are taken:

- **Tire pressure**
- **Tires** (are the correct tires mounted or are different types of tires mounted on the same axle).
- **Bearing play**
- **Unnecessary play** in kingpins, steering gear, and the balancing arms support on support axle vehicles.
- **The attachment of the axles** (play in spring-eye bolts, shackles, broken center rivets, broken springs and loose spring straps).
- **Worn torsion bar bushings**

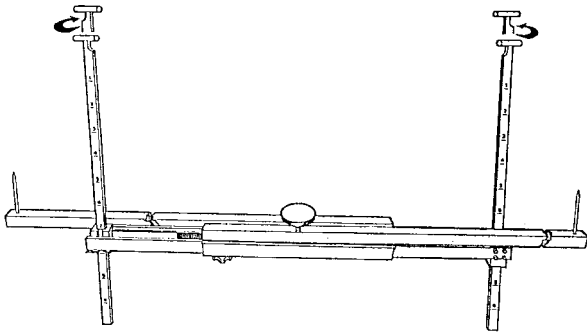
If any of the above mentioned points are not correct, these should be fixed before proceeding with the measurements.



Check of bearing play and bushings with JOSAM play detector AM 900:K

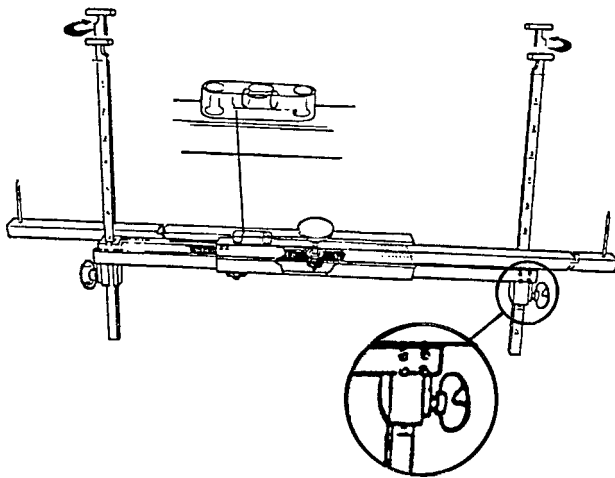


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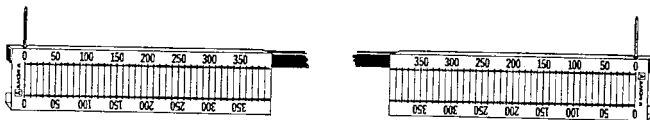


Frame gauge with hangers as well as gauge extensions with sight pins.

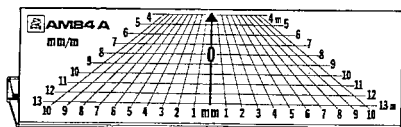
NOTE! The hangers are to be correctly placed with the straight side or cut side on the outside.



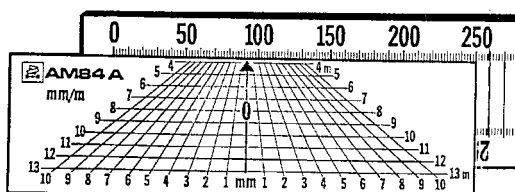
The frame gauge with stepless adjustable attachment for the hangers.
Gauge extension with bubble for checking the adjustment of horizontal for the frame gauges.



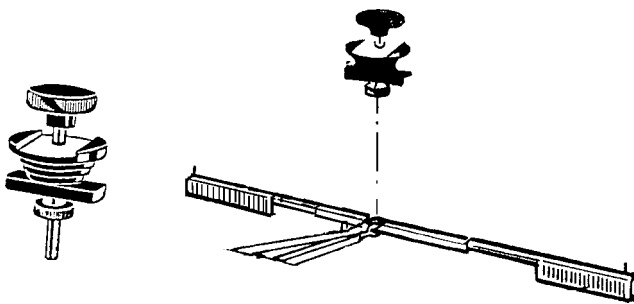
Measuring scale AM34 A.
Attach the gauge extension sight pin.



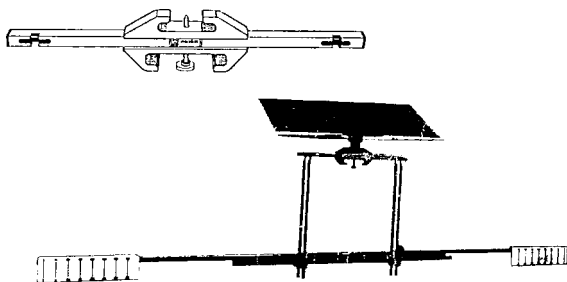
Sliding scale AM84 A.



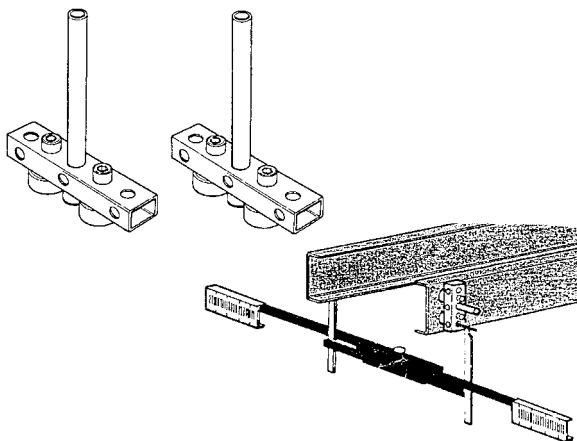
Sliding scale attached to the measuring scale.



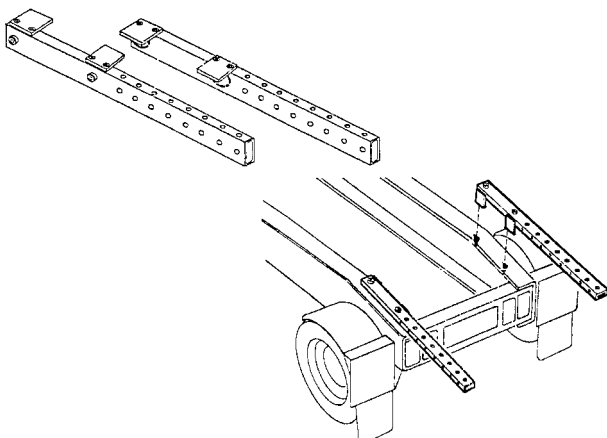
Trailer tow bar adapter AM48 A.
Attaches in the towing eyelet for gauge extensions
and measuring scales.



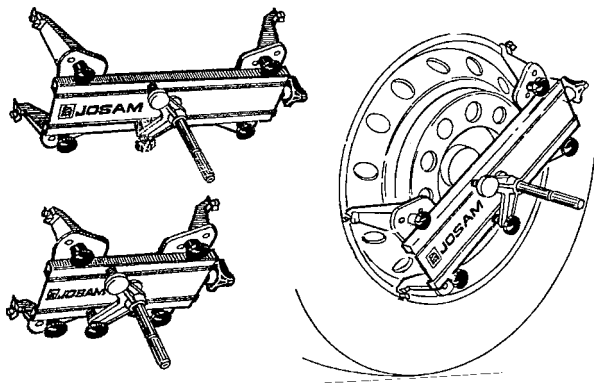
Adapter for kingpin AM69.
Attaches to kingpin for hanging the frame gauge.



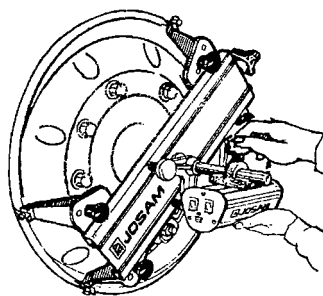
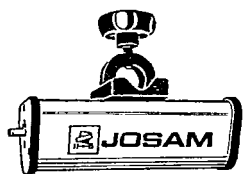
Magnet bracket for frame gauge AM172.
Attaches on sides of frame for hanging frame
gauges.



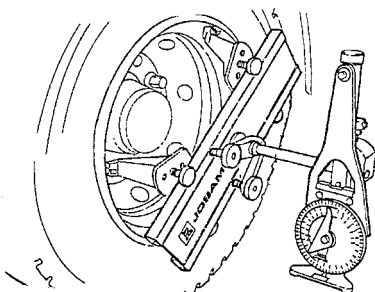
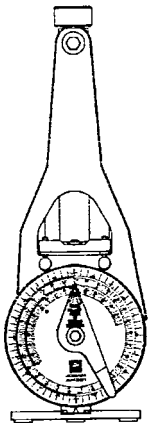
Magnet bracket for frame AM1227-1 and 2.
For hanging frame gauges on vehicles with short
overhang.



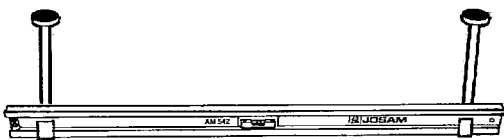
Wheel adapter AM10 and AM8.
AM10 for 16" – 24" wheel.
AM8 for 13" – 19" wheel.



Diode laser AM20-D1.
Battery-driven rechargeable laser projector.
Attaches to the axle of the wheel adapter.

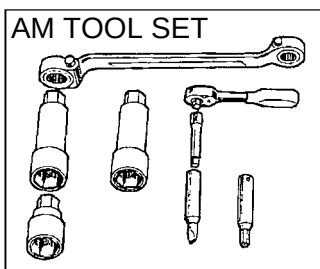


Wheel alignment gauge AM301.
Measuring instrument for camber, caster and KPI.
Attaches on the axle of the wheel adapter.

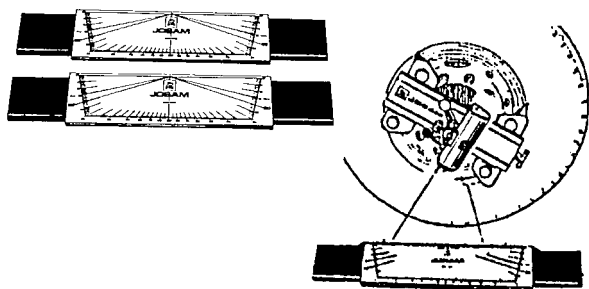


Spirit level AM542.
For checking of the horizontal on axle or frame.

Hanger AM98.
Magnetic hanger for the spirit level.

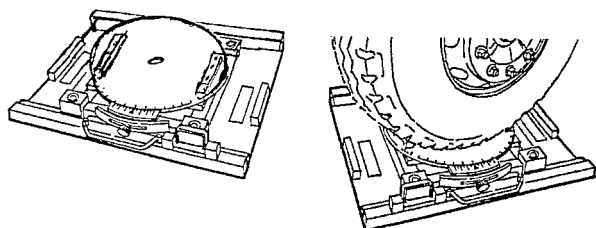


AM TOOL SET.
Tool set for the adjustment of the steering box.



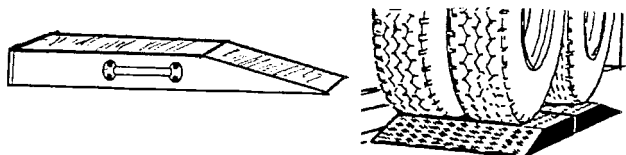
Turn angle gauge AM135.

Together with the laser projector measures maximum wheel turn, toe-out on turn, as well as position for KPI and caster.



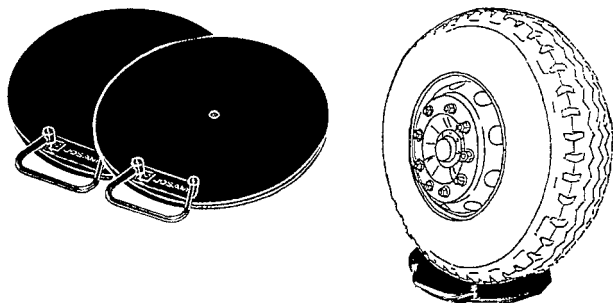
Turn plate AM585.

Designed to measure maximum turn, toe-out on turn, as well as KPI and caster.



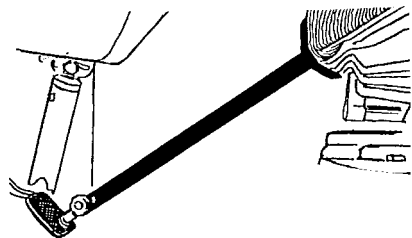
Compensation ramp AM931.

For example, compensation for height of the back wheel when turn plate AM585 is used.



Non-friction plate AM268.

To be placed under wheel when adjusting toe-in or toe-out.



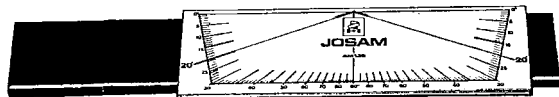
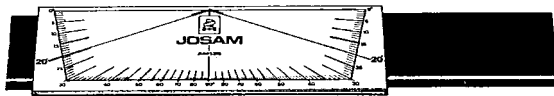
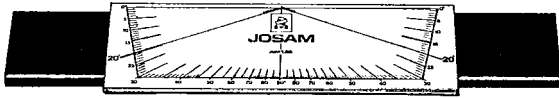
Brake rod AM325.

To be placed between the brake pedal and seat in order to brake the wheel during KPI measurement.

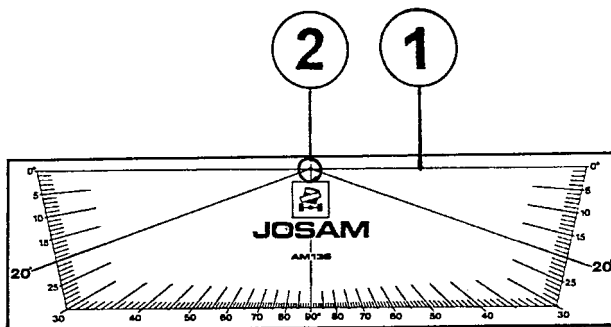


Description

Turn angle gauge AM135 is, together with the laser projector, designed to measure the maximum turn, toe-out on turn, and the fixed values for KPI and caster.

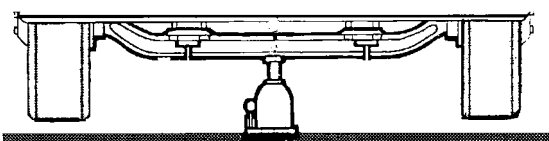


Measuring scales can slide along the base.

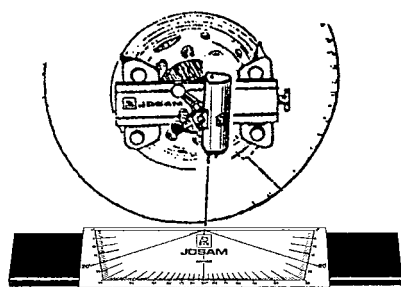


(1) The measuring scale's 0-line.

(2) The intersectional point of the 0-line.



For measurement using turn angle gauges AM135, the axles are lifted so that the wheel rolls freely approx. 10 mm from the floor. The caster value will be insignificantly altered.



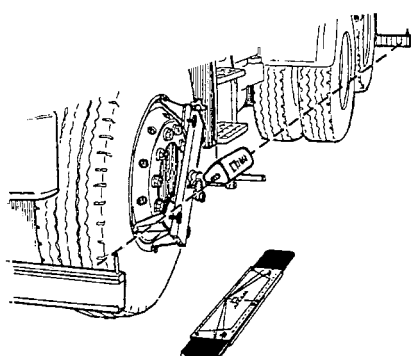
Hang frame gauge, gauge extensions and measuring scales.

Jack up the axle.

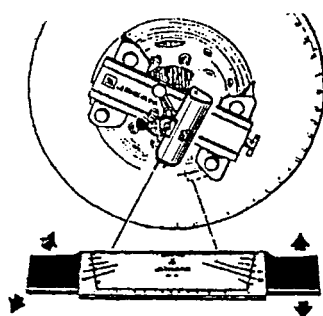
Attach wheel adapters and laser projector.

Run-out of the wheel can be compensated with the adjustment knobs on the adapter

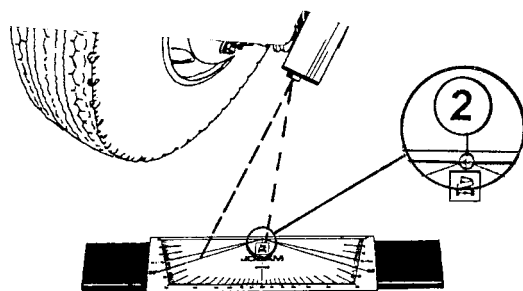
Place the turn angle gauges on the floor for both the left and right sides, with a centered scale and in the middle of the wheel.



Position left, and alternatively, right wheel in a straight ahead position with the laser projector. Equal values on the front and rear measuring scales.



Align turn angle gauges on both sides with laser dot on the 0-line of the measuring scale. Move measuring scale along the profile and check at both ends that the laser dot hits the 0-line.



Turn wheel

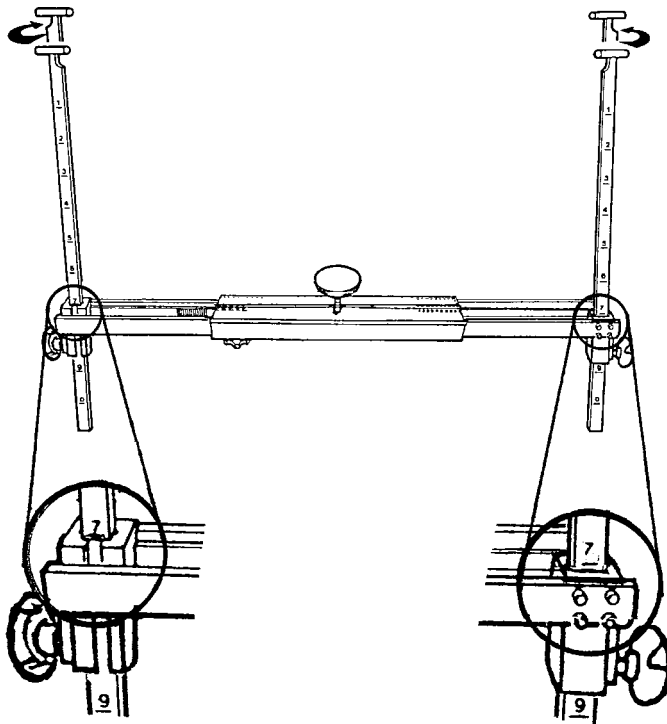
Align the laser dot on the 0-line **(1)** of the measuring scale.

Move measuring scale until laser dot hits the intersectional point of the 0-line **(2)**.

Align the laser dot on the degree scale and read the value.

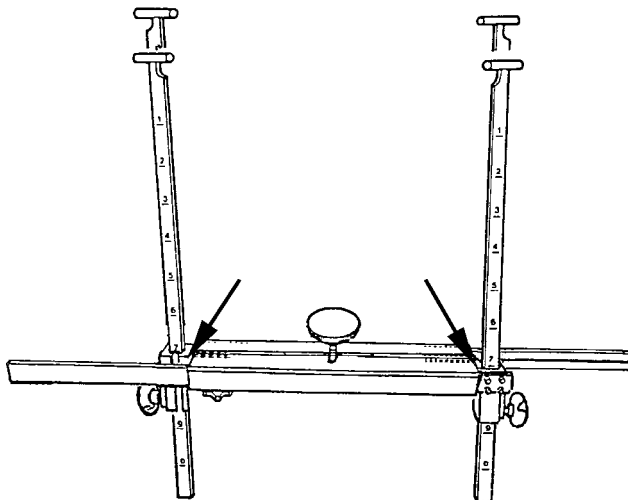


Checking the frame gauge



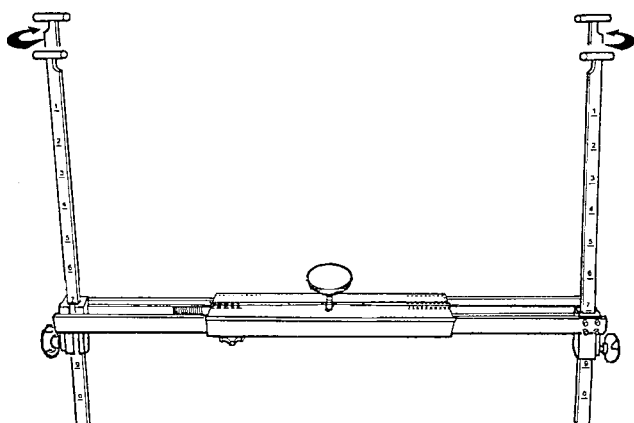
Mount hangers at both ends.

Both hangers should be placed with the straight side or the cut side facing out. They should hang at the same height. Check that the numerical marks are aligned with the upper edge of the end piece. Lock with the knob.



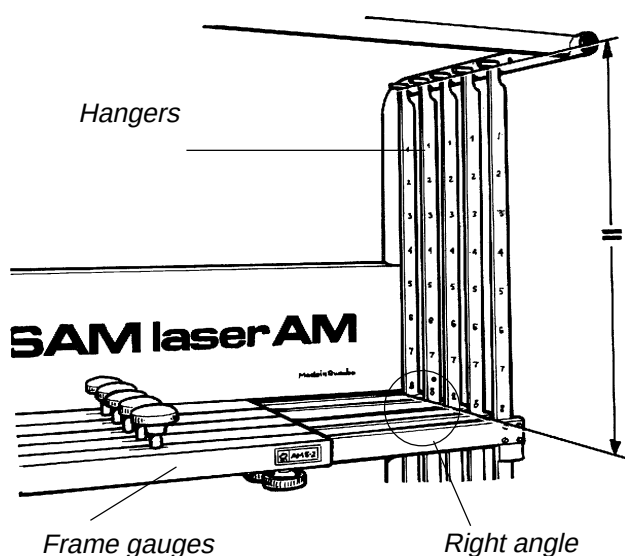
Move together the guide rails on the frame gauge. Both ends should face toward the housing.

Check that the guide rails are easy to move back and forth on the housing.



Checking the frame gauge

The self-centering frame gauges must be checked while they are hanging in the storage trolley.

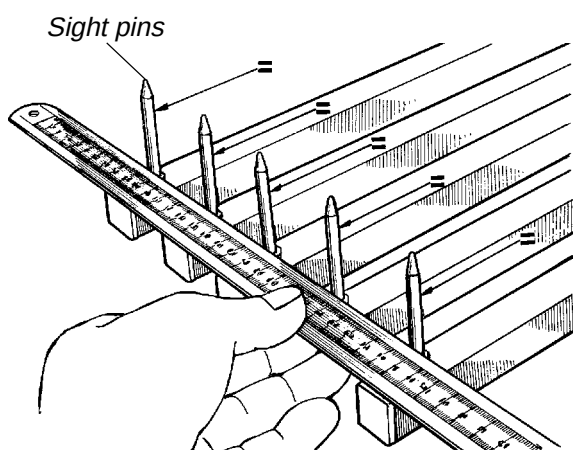


Hang the frame gauges on the support rack in the storage trolley.

At least three frame gauges are required.

Make sure that the hangers are hanging at the same height and are all uniformly facing forward and tightly fixed in towards the support rack.

Check also that the frame gauges hang at the same height, and that there is a right angle between the frame gauges and the hangers.

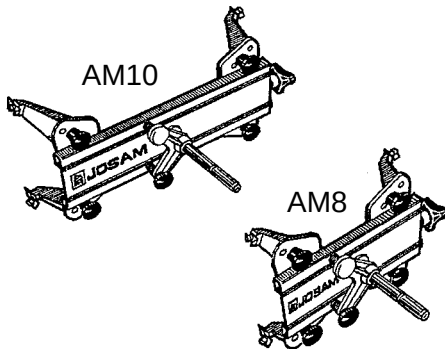


Mount the gauge extensions.
Check the sight pins with a straight edge.

They should lie in a line.

The maximum allowable deviation is 2 millimeters.

Turn a frame gauge 180° and check the sight pins again. If the deviation exceeds the tolerance level, then one or more of the hangers are bent.



Wheel adapter AM10

Intended for rims between 16" – 24".
The adapter is equipped with reversible grip hooks.

Wheel adapter AM8

Intended for rims between 13" – 19".
The adapter is equipped with reversible grip hooks.

Attachment of the wheel adapter

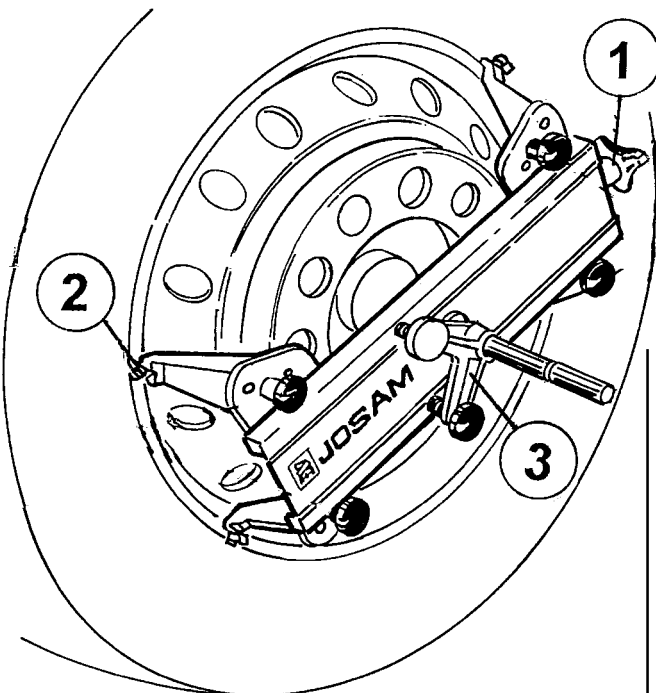
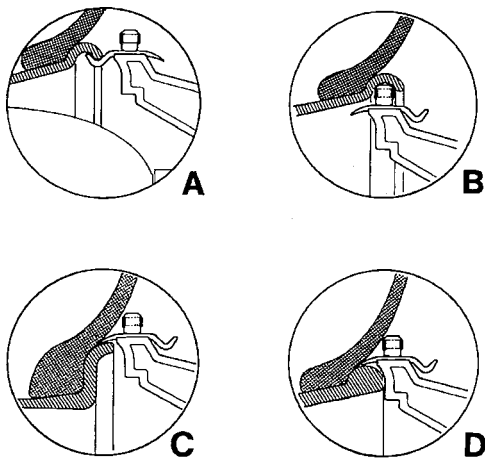
Attachment is done with the knob **(1)** on the short side of the adapter.

On rims with corners, the wheel adapter is attached according to figures **A** and **B** and tightened outwards.

On the rims shown in figure **C** or on aluminium rims like figure **D**, the grip hooks are reversed and attached between the tire and the rim and tightened inwards.

During the clamping process, check that the grip hooks are gripping properly on the rim edges (figures **A** and **B**) or gripping in between the tire and the rim (figures **C** and **D**).

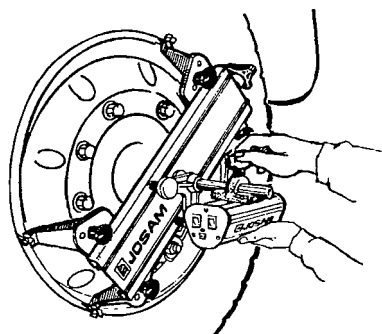
Press the adapter against the rim during the clamping with the knob **(1)**.



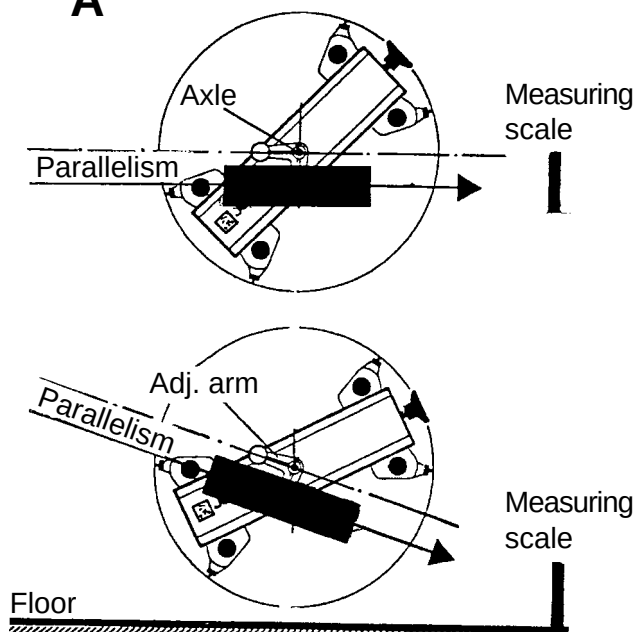
(1) Knob for clamping

(2) Grip hooks

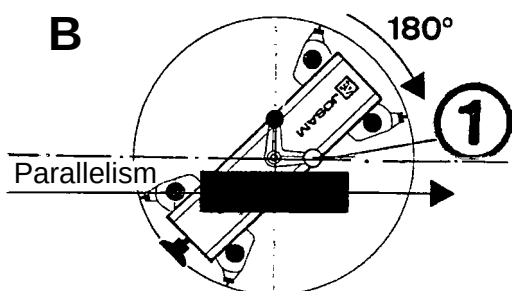
(3) Adjustment arm



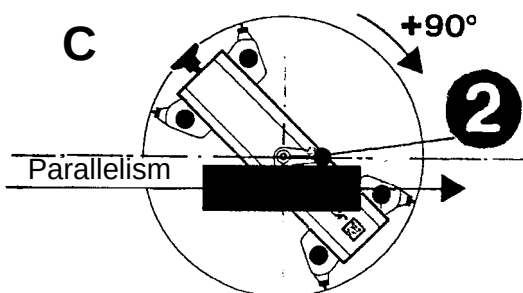
A



B



C



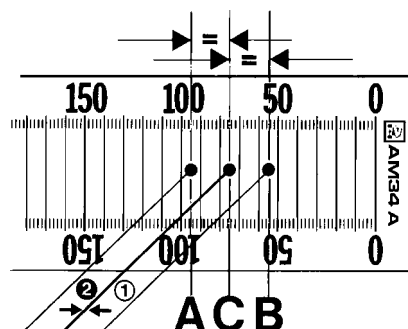
Hang frame gauges, gauge extensions, and measuring scales. Jack up the axle so that you can spin the wheel. Attach the wheel adapter on the wheel (wheels). Attach the laser projector on the axle of the wheel adapter. The guide pin should run in the groove. Turn the knob so that rotating the laser projector meets with some resistance.

The measuring scales are to be placed 2–4 meters from the laser projector. The measuring scales can be placed on the floor if the distance is less than 2 meters or more than 4 meters. When the distance is too short or too long between the measuring scales and the laser projector, the measurements are not as exact.

A Place the laser projector parallel with the adjustment arm. Rotate the wheel and aim the laser dot at the measuring scale.

Read the indicated value (for example 95)

Value A. Write down the result.



B Rotate the wheel a half turn (180°) and aim the laser dot at the measuring scale.

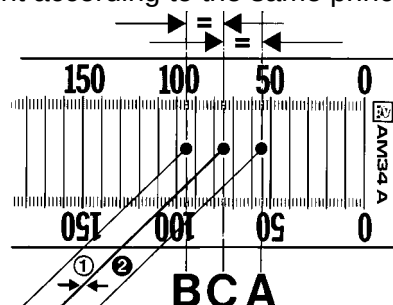
Read the indicated value (for example 55)

Value B.

Adjust with the “white” knob (1) so that the laser dot hits right between **A** and **B** at **C**. The value is 75 in this example.

C Rotate the wheel a quarter turn (90°). Adjust with the “black” knob (2) until the laser dot hits the earlier value **C75**. Rotate the wheel and check to make sure the laser dot has not “wandered” to the side. Repeat the adjustment if the laser dot “wanders”.

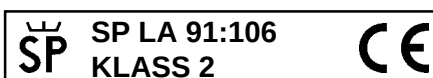
If value **A** is less than value **B** carry out the adjustment according to the same principle.





JOSAM laser AM Measuring system

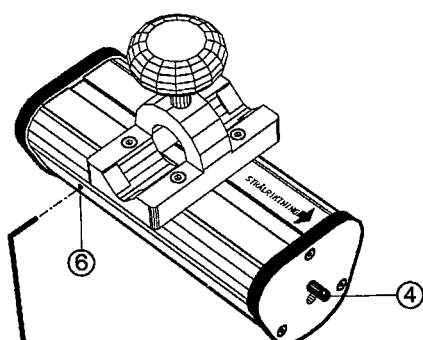
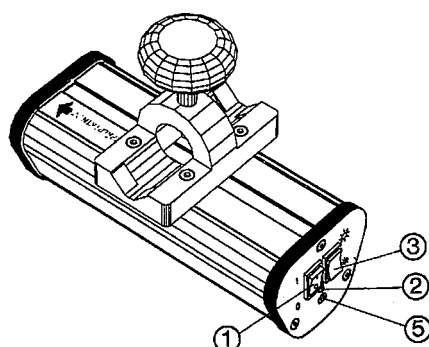
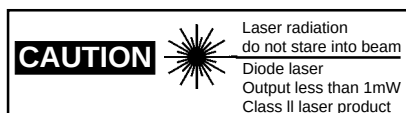
Laser projector AM20-D 1



TYPE: AM20-D 1 MADE IN SWEDEN

SERIAL NO _____

MANUFACTURED _____



Description

The laser is powered by 6 rechargeable R 14 Ni-Cad batteries.

Charging takes place with a power operated transformer AM22-D 1 connected to the contact (5) on the rear end.

The normal charging time is 14 hours.

Operation time is approx. 50 hours.

Change of batteries

- Remove the front end, 3 screws.
- Replace the batteries.
- NOTE! Place the batteries according to the diagram in the housing.
- Replace the front end on.

Respect the text on the warning labels.

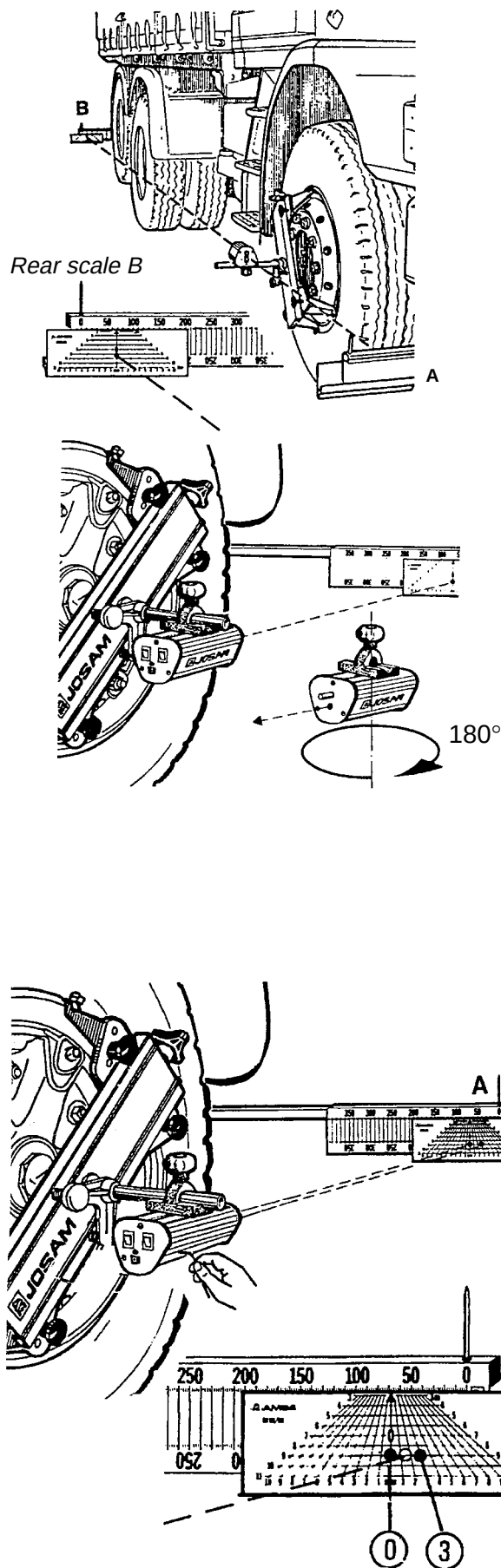
The manufacturer will not be held responsible if the laser projector is used for any other purpose than that for which it is intended.

Function

- (1) The on/off circuit breaker.
- (2) The light indicates that the laser is on.
When the laser is turned on give it full light intensity. After approx. 10 minutes reduce the light intensity by one half.
- (3) A circuit breaker is used for regulation of light intensity.
For full light intensity press on the upper part of the circuit breaker, for reduced light intensity, press on the bottom part of the circuit breaker.
- (4) The shutter.
The opening for the laser beam can be closed by turning the knob a half turn.

Adjustment

The angle of the laser beam in relation to the wheel with the wheel adapter is adjusted by turning an adjustment screw (6), located on the side of the housing, with a 2 mm Allen wrench.



Calibration of laser projector

Hang the frame gauges front and back, with measuring scales AM34 A and sliding scales AM84 A.

Attach the wheel adapter and laser projector.

Aim the laser dot at the front measuring scale **(A)**. Move the sliding scale so that the laser dot hits the O-line.

Rotate the laser projector on the axle with the wheel adapter.

Align the laser dot on the rear measuring scale **(B)**. Move the sliding scale so that the laser dot hits the O-line.

Detach the laser projector and **turn** it around 180° on the axle of the wheel adapter.

Align the laser dot on the O-line of the front and rear sliding scales.

The laser dot should hit the O-line on both scales.

**Max. allowable deviation:
1 mm per 6 meters measured distance**

If the laser dot hits at an equal distance from the O-line on the outside of both or on the inside of both scales; then the laser beam is at the correct angle, but has a parallel deviance that can be tolerated.

Example: If the laser dot hits distance line 3 (after you have turned the laser upside down) adjust the laser dot.

With a 2 mm Allen wrench, adjust the adjusting screw on the laser housing located according to (6) on page 14, to a value 1.5 on the same distance line (the value halfway between 0 and 3).

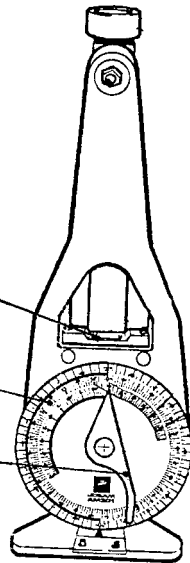
Check again that the same value is maintained front and rear by putting the sliding scales on 0 and **turning** the laser projector 180° on the axle of the wheel adapter.



Measuring level

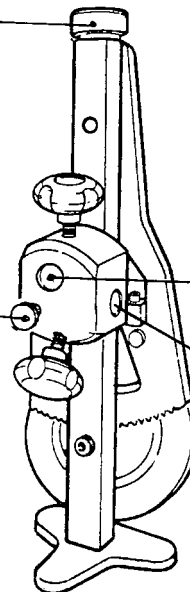
Measuring Scale

Indicator



Upper level

Lock



Measuring instrument for camber, caster and KPI.

Description

The wheel alignment gauge AM301 has a rotating scale and an adjustable measuring spirit level.

The scale is divided into degrees and minutes, minus values (-) are marked in red.

On the outer part of the scale, camber can be read by looking at the arrow on the base of the scale.

On the inner part of the scale, caster and KPI can be read on the adjustable indicator.

The top bubble is for adjusting the vertical position.

The wheel alignment gauge should be attached to the wheel adapter spindle.

Attachment hole for KPI measurements.

Move the instrument towards the bottom and lock with the upper knob.

Attachment hole for measuring of camber and caster.

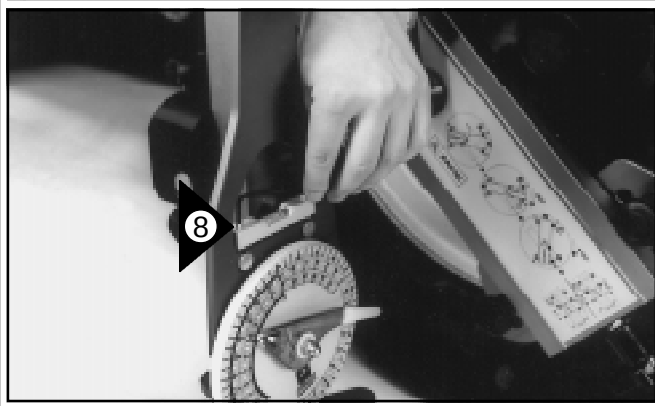
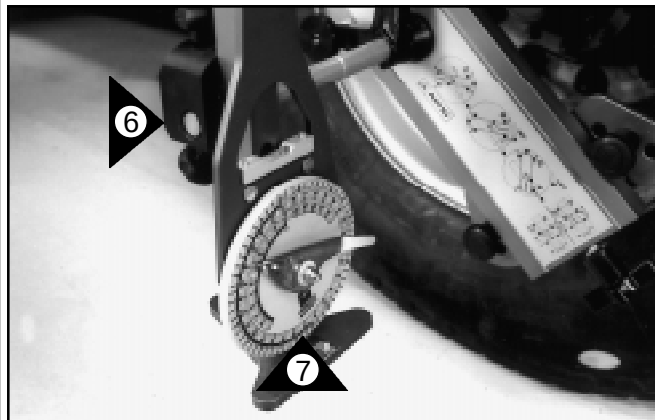
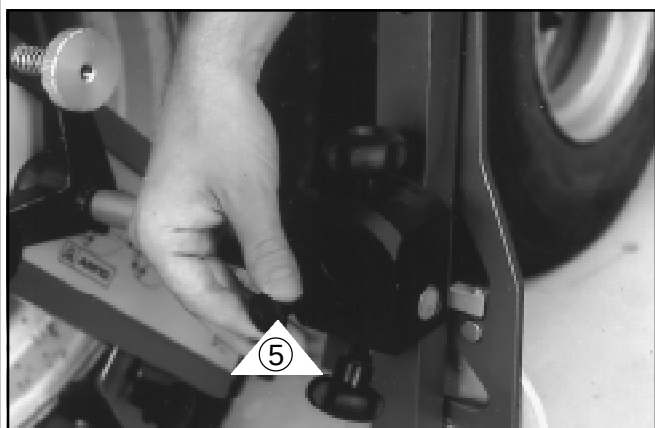
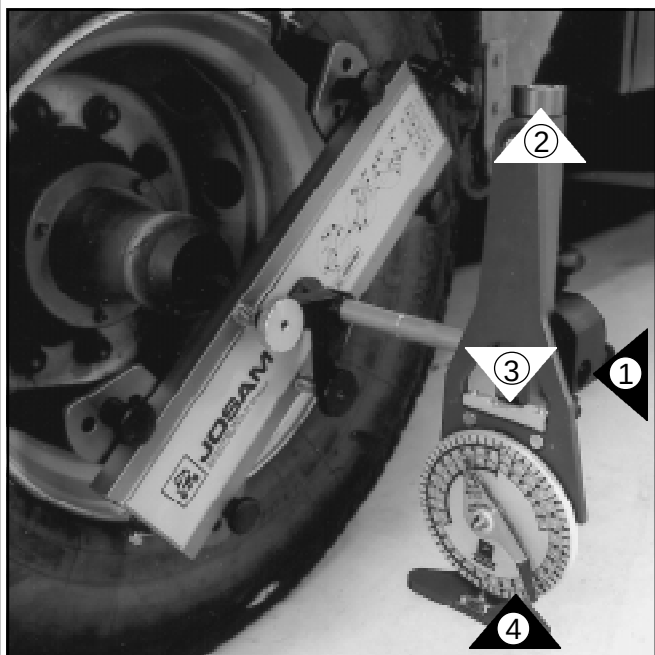
Move the instrument towards the locking pin and lock with the bottom knob.



AM301 for measurement of camber/caster



AM301 for KPI measurements



Calibration

Mount the wheel adapter and compensate for run-out of the wheel with the adjustment knobs on the adapter.

(1) Mount the AM301 to the wheel adapter by inserting the adapter spindle into the hole for camber and caster measuring. Move the instrument towards the locking pin.

(2) Check the top bubble, and lock with the lower knob.

(3) Turn the measuring scale until the measuring bubble shows the horizontal position.

(4) Read the camber from the outer numbers on the scale. Record the values.

NOTE! Do not forget to write down if it is a (+) or a (-) number.

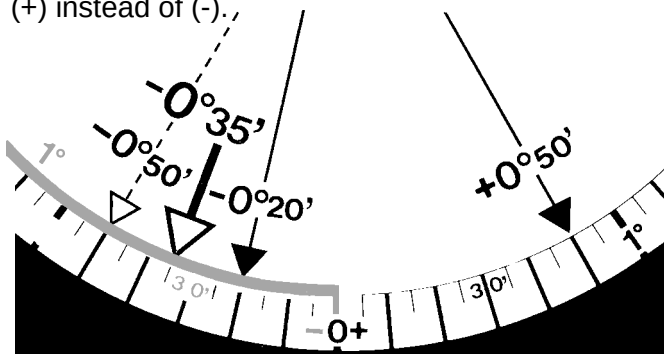
Example: $+0^{\circ} 50'$

(5) Take AM301 off the wheel adapter spindle.

(6) Mount it the opposite way by pulling out the locking pin.

(7) Turn the measuring scale so that the measuring bubble shows a horizontal position. Read the camber value.

If $-0^{\circ} 50'$ is the reading, the alignment gauge is correctly calibrated. This is the same value as the first measurement but with the opposite sign (i.e. (+) instead of (-)).



If different numbers come up, the measuring level has to be adjusted.

Example:

Reading in the first case = $+0^{\circ} 50'$ camber.

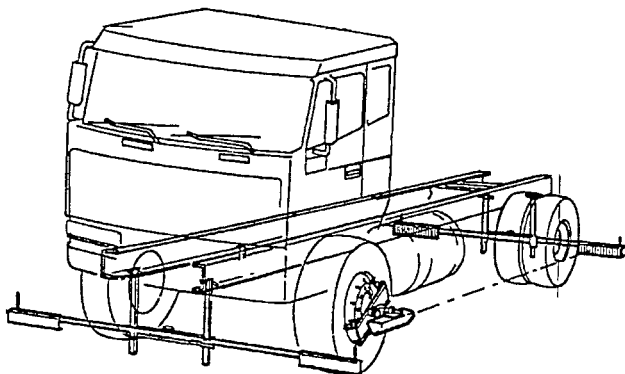
Reading in the second case = $-0^{\circ} 20'$ camber.

The second reading should have been $-0^{\circ} 50'$.

Adjust the measuring scale to $-0^{\circ} 35'$ camber, which is halfway between $-0^{\circ} 50'$ and $-0^{\circ} 20'$.

(8) Adjust the measuring bubble with one of the screws until the measuring bubble shows it is horizontal.

Check the numbers by repeating the first measurement.



Hanging the frame gauges on a truck

Hang the frame gauges with hangers and gauge extensions as far forward and as far back as possible on the frame.

Adjust the height on the hangers so that the frame gauge and hangers are at the same height as the center of the wheel.

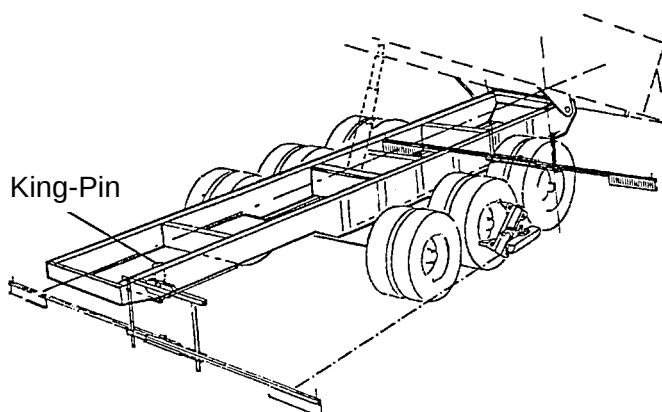
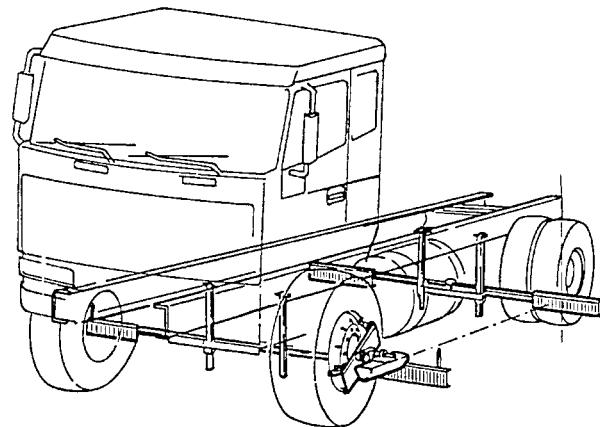
Place the measuring scales on the gauge extension sight pins.

Adjust the frame gauge with stepless adjustable mounting for the hangers and gauge extension with bubble to horizontal position.

NOTE! Make sure during the installation of the frame gauge that the attachment on the truck frame flange is the same on both sides.

Check that the hangers are at a right angle to the frame by comparing them with the cross-member or spring mounting.

If it is impossible to hang the frame gauges all the way forward, or all the way back, then these can be hung behind the front axle and in front of the rear axle.



Hanging the frame gauges on a semi-trailer

Mount the adapter AM69 on the kingpin and hang the frame gauge on the adapter.

Check that the frame gauge and the gauge extensions are at a right angle to the frame by:

Aiming the laser dot at the measuring scale. Turn the frame gauge, gauge extensions and adapters on the kingpin and observe the laser dot on the measuring scale.

When the laser dot shows the **HIGHEST** value, the frame gauge and gauge extensions are at a right angle to the frame.

Hang the rear frame gauge as far back as possible.



Hanging frame gauges on a trailer with a trailer hitch

Front axle(s) on "dolly"

Attach trailer tow bar adapter AM48 in the towing eyelet.

Attach the gauge extensions to the trailer tow bar adapter and place the measuring scales on the sight pins.

Check that the gauge extensions are at a right angle to the tow bar by:

Aligning the laser dot on the measuring scale.

Turn the front gauge extensions connected by the AM48 and observe the laser dot on the measuring scale.

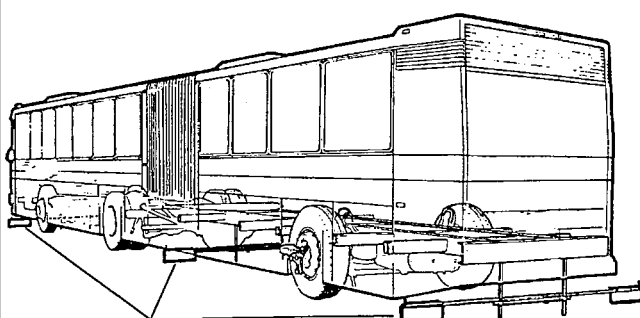
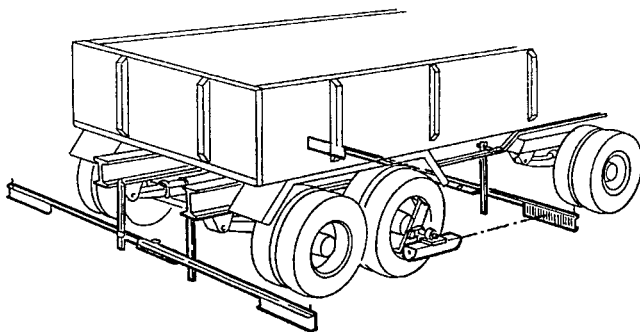
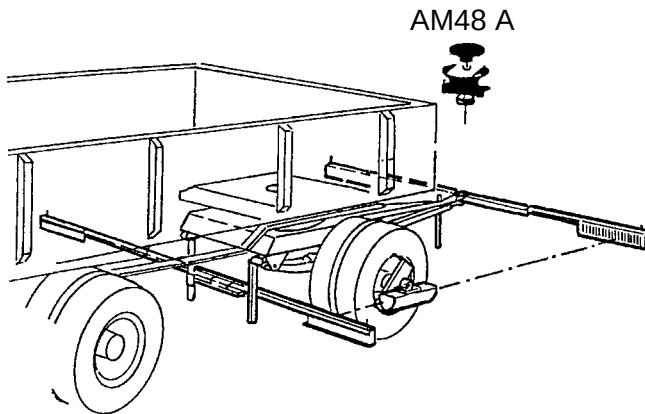
When the laser dot shows the **HIGHEST** value, the gauge extensions are at a right angle to the tow bar. Hang the rear frame gauge to the rear end of the front axle frame (the dolly).

Rear axle(s)

Hang the frame gauges front and rear on the frame side members of the truck frame. Check that they hang at a right angle to the truck frame by comparing them with the crossbars or the spring mounting.

Hanging frame gauges on a bus with steerable rear axle

Requires 3 frame gauges. Hang two frame gauges on the front section of the bus- one all the way forward and one all the way back. Hang a frame gauge on the back section of the bus at the extreme rear.

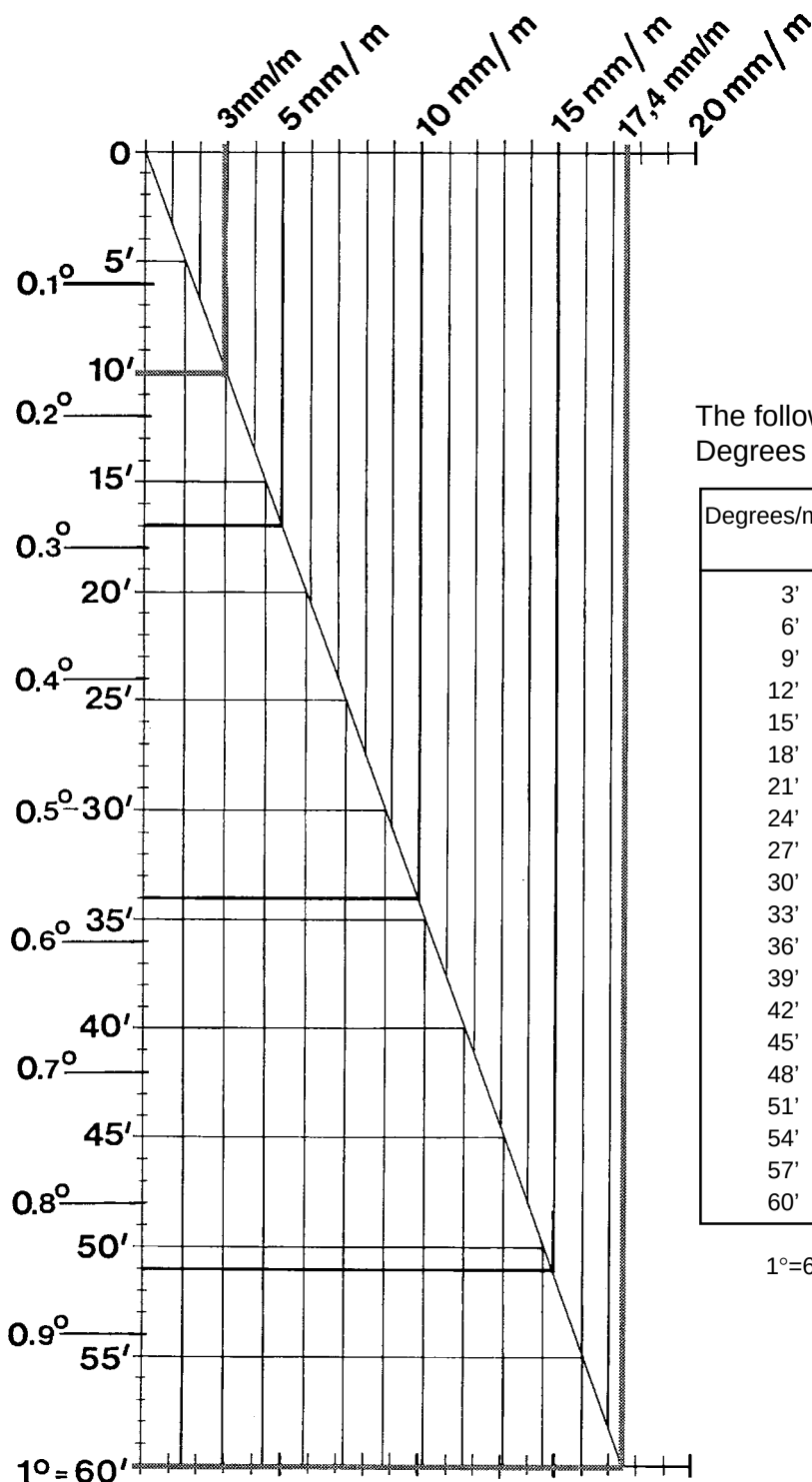


Frame gauges and measuring scales



Table mm/m compared with degrees and minutes

If the manufacturer gives tolerances in degrees and minutes this table can be used for converting to mm/m. For example: $10' = 3 \text{ mm/m}$. $1^\circ = 17.4 \text{ mm/m}$.



The following table shows comparison values.
Degrees in tenths and mm/m

Degrees/minutes	Degrees/ in tenths	mm/m
3'	0.05°	0.9
6'	0.10°	1.7
9'	0.15°	2.6
12'	0.20°	3.5
15'	0.25°	4.4
18'	0.30°	5.2
21'	0.35°	6.1
24'	0.40°	7.0
27'	0.45°	7.8
30'	0.50°	8.7
33'	0.55°	9.6
36'	0.60°	10.5
39'	0.65°	11.3
42'	0.70°	12.3
45'	0.75°	13.1
48'	0.80°	14.0
51'	0.85°	14.8
54'	0.90°	15.7
57'	0.95°	16.6
60'	1.00°	17.4

$1^\circ = 60'$

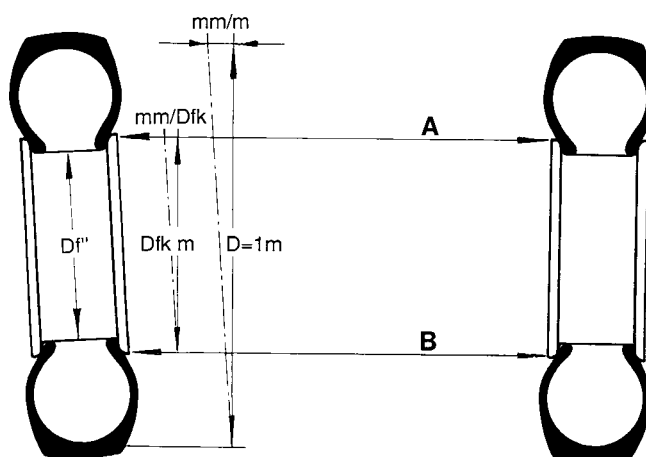


*) tubeless

Table mm/m, tolerance in mm per bead of rim

C/D	F/Dfk mm/m										Df''	Dfk m
	24"	*) 22 1/2", 20"	19"	18"	17"	16"	*) 17 1/2", 15"	*) 14"	*) 13"	*) 12"		
	mm/m	0,68	0,58	0,55	0,525	0,50	0,475	0,45	0,385	0,36	0,335	
0,5	0,34	0,29	0,28	0,26	0,25	0,24	0,23	0,19	0,18	0,17		
1	0,68	0,58	0,55	0,53	0,50	0,48	0,45	0,39	0,36	0,34		
1,5	1	0,9	0,8	0,8	0,8	0,7	0,7	0,6	0,5	0,5		
2	1,4	1,2	1,1	1,1	1	1	0,9	0,8	0,7	0,7		
2,5	1,7	1,5	1,4	1,3	1,3	1,2	1,1	1	0,9	0,8		
3	2	1,7	1,7	1,6	1,5	1,4	1,4	1,2	1,1	1		
3,5	2,4	2	1,9	1,8	1,8	1,7	1,6	1,4	1,3	1,2		
4	2,7	2,3	2,2	2,1	2	1,9	1,8	1,5	1,4	1,3		
4,5	3,1	2,6	2,5	2,4	2,3	2,1	2	1,7	1,6	1,5		
5	3,4	2,9	2,8	2,6	2,5	2,4	2,3	1,9	1,8	1,7		
5,5	3,7	3,2	3	2,9	2,8	2,6	2,5	2,1	2	1,8		
6	4,1	3,5	3,3	3,2	3	2,9	2,7	2,3	2,2	2		
6,5	4,4	3,8	3,6	3,4	3,3	3,1	2,9	2,5	2,3	2,2		
7	4,8	4,1	3,9	3,7	3,5	3,3	3,2	2,7	2,5	2,4		
7,5	5,1	4,4	4,1	3,9	3,8	3,6	3,4	2,9	2,7	2,5		
8	5,4	4,6	4,4	4,2	4	3,8	3,6	3,1	2,9	2,7		
8,5	5,8	4,9	4,7	4,5	4,3	4	3,8	3,3	3,1	2,9		
9	6,1	5,2	5	4,7	4,5	4,3	4,1	3,5	3,2	3		
9,5	6,5	5,5	5,2	5	4,8	4,5	4,3	3,7	3,4	3,2		
10	6,8	5,8	5,5	5,3	5	4,8	4,5	3,9	3,6	3,4		
10,5	7,1	6,1	5,8	5,5	5,3	5	4,7	4	3,8	3,5		
11	7,5	6,4	6,1	5,8	5,5	5,2	5	4,2	4	3,7		
11,5	7,8	6,7	6,3	6	5,8	5,5	5,2	4,4	4,1	3,9		
12	8,2	7	6,6	6,3	6	5,7	5,4	4,6	4,3	4		
12,5	8,5	7,3	6,9	6,6	6,3	5,9	5,6	4,8	4,5	4,2		
13	8,8	7,5	7,2	6,8	6,5	6,2	5,9	5	4,7	4,4		
13,5	9,2	7,8	7,4	7,1	6,8	6,4	6,1	5,2	4,9	4,5		
14	9,5	8,1	7,7	7,4	7	6,7	6,3	5,4	5	4,7		
14,5	9,9	8,4	8	7,6	7,3	6,9	6,5	5,6	5,2	4,9		
15	10,2	8,7	8,3	7,9	7,5	7,1	6,8	5,8	5,4	5		
15,5		9	8,5	8,1	7,8	7,4	7	6	5,6	5,2		
16		9,3	8,8	8,4	8	7,6	7,2	6,2	5,8	5,4		
16,5		9,6	9,1	8,7	8,3	7,8	7,4	6,4	5,9	5,5		
17		9,9	9,4	8,9	8,5	8,1	7,7	6,6	6,1	5,7		
17,5		10,2	9,6	9,2	8,8	8,3	7,9	6,7	6,3	5,9		
18			9,9	9,5	9	8,6	8,1	6,9	6,5	6		
18,5			10,2	9,7	9,3	8,8	8,3	7,1	6,7	6,2		
19				10	9,5	9	8,6	7,3	6,8	6,4		
19,5					9,8	9,3	8,8	7,5	7	6,5		
20					10	9,5	9	7,7	7,2	6,7		
20,5						9,7	9,2	7,9	7,4	6,9		
21						10	9,5	8,1	7,6	7		
21,5							9,7	8,3	7,7	7,2		
22							9,9	8,5	7,9	7,4		
22,5							10,1	8,7	8,1	7,5		
23								8,9	8,3	7,7		
23,5								9,1	8,5	7,9		
24								9,2	8,6	8		
24,5								9,4	8,8	8,2		
25								9,6	9	8,4		
25,5								9,8	9,2	8,5		
26								10	9,4	8,7		
26,5									9,5	8,9		
27									9,7	9,1		
27,5									9,9	9,2		
28									10,1	9,4		
28,5										9,6		
29										9,7		
29,5										9,9		
30										10,1		

Tolerance in mm per bead of rim



A-B = mm Ø bead of rim

In certain cases, the manufacturer gives tolerances in mm per bead of rim (in the middle of the bead of rim). Compare with mm/m.



The complete Test Report Sheet AM39



JOSAM ©
Box 419 · S-701 48 ÖREBRO · SWEDEN

Reg. n° _____ Type _____
Date _____ Name _____
Miles/Km _____ Sign _____

KURVINKELDIFFERENS
SPURDIFFERENSWINKEL
TOE-OUT ON TURN
DIFFERENCE DE COURBE D'ANGLE

VÄNSTER
LINKS
LEFT
GAUCHE

HÖGER
RECHTS
RIGHT
DROIT

DIFF.

20°

MAX. SVÄNG
MAX. LENKEINSCHLAG
MAX. TURN
BRAQUAGE MAX.

HÖGER
RECHTS
RIGHT
DROIT

20°

VÄNSTER
LINKS
LEFT
GAUCHE

DIFF.

A1

B1

C1 ± : D_m =

CAMBER
STURZ
CARROSSAGE
+

KPI
Spreizung
INCLIN.
PIVOTS
+

CASTER
NACHLAUF
CHASSE
+

A2

B2

C2 ± : D_m =

CAMBER
STURZ
CARROSSAGE
+

KPI
Spreizung
INCLIN.
PIVOTS
+

CASTER
NACHLAUF
CHASSE
+

A1

B1

C3 ± : D_m =

CAMBER
STURZ
CARROSSAGE
+

A2

B2

C4 ± : D_m =

CAMBER
STURZ
CARROSSAGE
+

A1

B1

C5 ± : D_m =

CAMBER
STURZ
CARROSSAGE
+

A2

B2

C6 ± : D_m =

CAMBER
STURZ
CARROSSAGE
+

B1

B2

SNEDSTÄLLNING
SCHRAGSTAND
OUT OF SQUARE
ANGLE FAUSSE

DIFF C₃/C₄-C₅/C₆

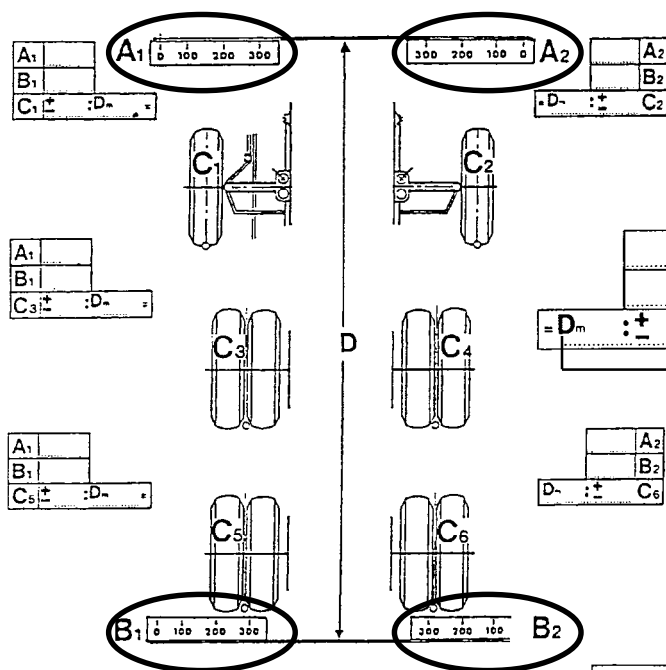
C₄ ±

C₆ ±

A_{1.2}

B_{1.2}

= D_m

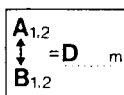


Front frame gauge with measuring scale
A1 Left **A2 Right**

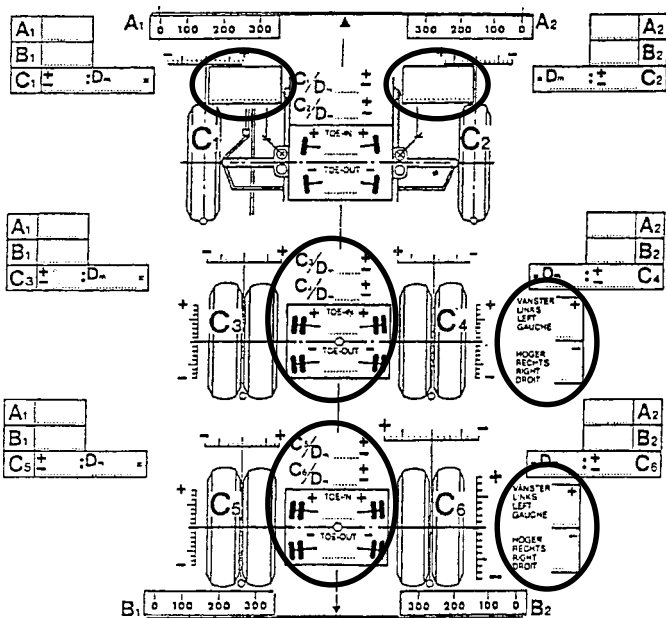
A_1 — Value from front measuring scale **A** in mm
 B_1 — Value from rear measuring scale **B** in mm
 C_1 — C = Calculated value between A and B
 D — D = Distance between scales in meters



Rear frame gauge with measuring scales
B1 Left **B2 Right**



Measured distance between scales A and B



Measured value of location of front axle,
out of square position

C_3/D_m — Rolling direction of left wheel mm/m
 C_4/D_m — Rolling direction of right wheel mm/m

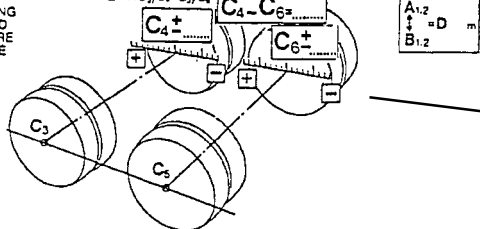
Calculated toe-in/toe-out mm/m

VANSTER
LINKS
LEFT
GAUCHE
HÖGER
RECHTS
RIGHT
DROIT

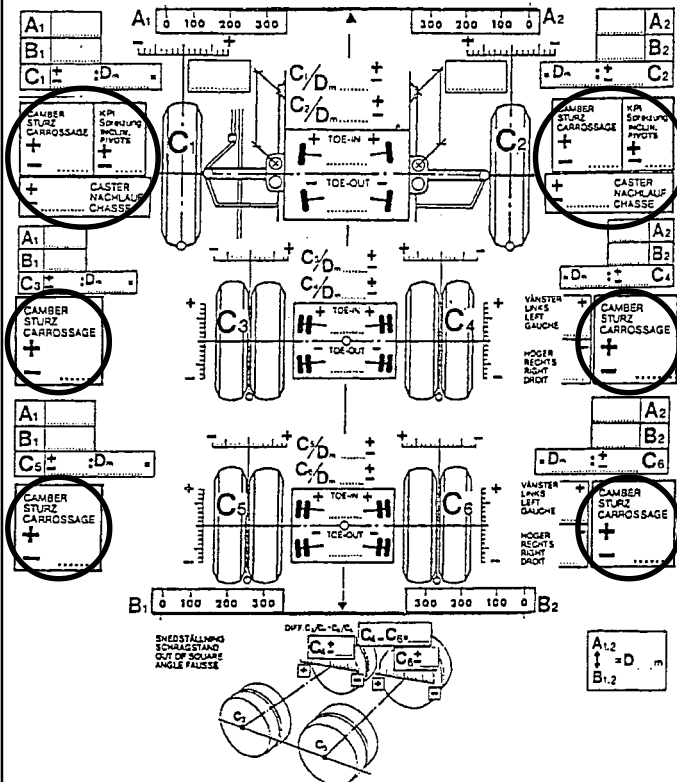
Rolling direction of wheel, out of
square position of rear axle(s)
Left + and Right -

SNEDSTÄLLNING
SCHRAGSTAND
OUT OF SQUARE
ANGLE FAUSSE

DIFF. $C_3/C_4 - C_5/C_6$



Calculation of out of square and
parallelism between the axles

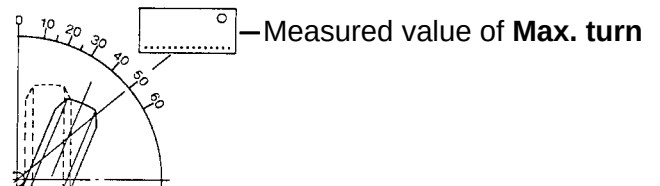
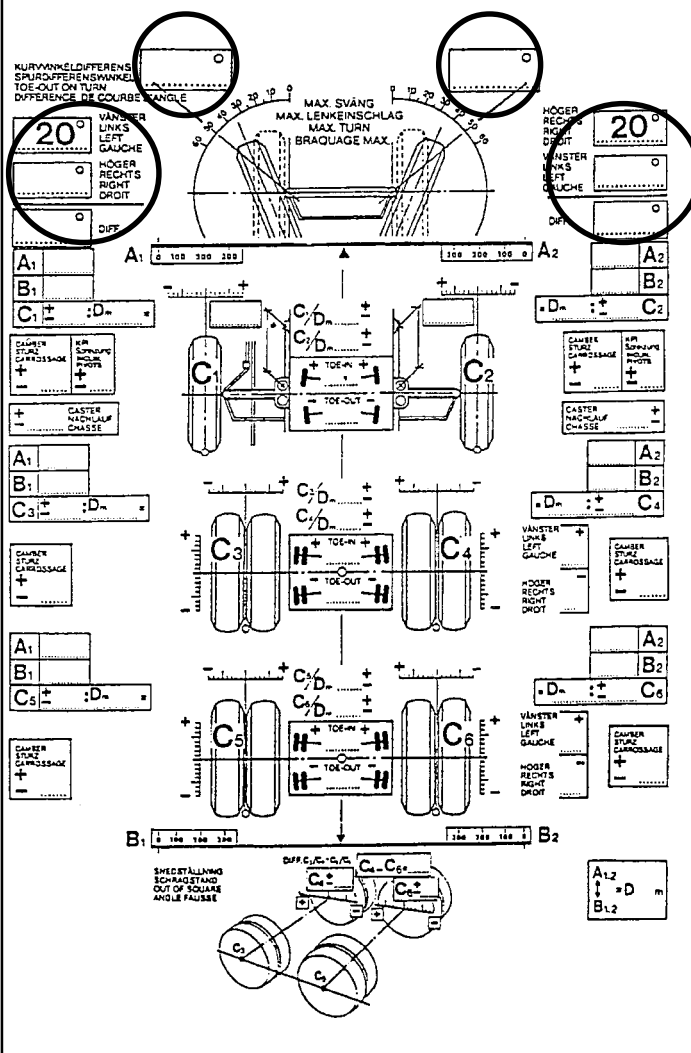


Measured values of
the front axle:

CAMBER STURZ CARROSSAGE	KPI Spreizung INCLIN. PIVOTS
+	+
-	-
CASTER NACHLAUF CHASSE	
+	
-	

Measured values of rear
axle(s):

CAMBER STURZ CARROSSAGE
+
-

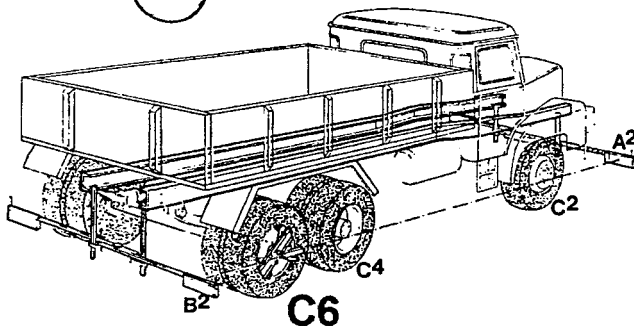
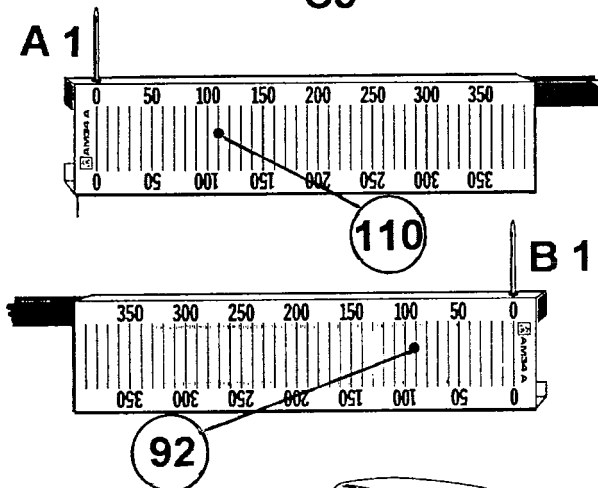
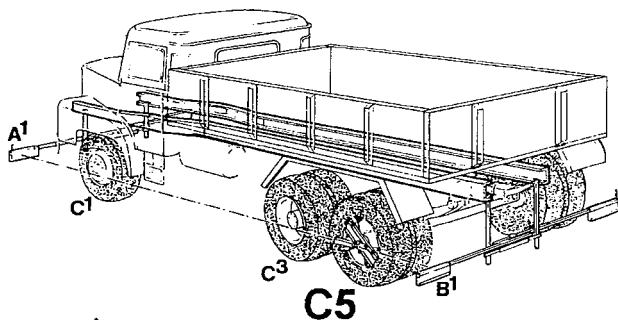


HÖGER RECHTS RIGHT DROIT	20°	Output value
VANSTER LINKS LEFT GAUCHE		Measured value on opposite side
DIFF.		Toe-out on turn



JOSAM laser AM Measuring system

Measurement of rigid axles



Total difference between front and rear measuring scales
Distance in meters between the measuring scales

C/D		mm m															
C _{mm}	D _m	3	3,5	4	4,5	5	5,5	6	6,5	7	7,5	8	8,5	9			
		3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
3	1	0,9	0,8	0,7	0,6	0,5	0,4	0,3	0,2	0,1	0,0	0,0	0,0	0,0	0,0	0,0	0,0
4	1,3	1,1	1,0	0,9	0,8	0,7	0,6	0,5	0,4	0,3	0,2	0,1	0,0	0,0	0,0	0,0	0,0
5	1,7	1,4	1,3	1,1	1,0	0,9	0,8	0,7	0,6	0,5	0,4	0,3	0,2	0,1	0,0	0,0	0,0
6	2,1	1,7	1,5	1,3	1,2	1,1	1,0	0,9	0,8	0,7	0,6	0,5	0,4	0,3	0,2	0,1	0,0
7	2,5	2,1	1,8	1,6	1,4	1,3	1,2	1,1	1,0	0,9	0,8	0,7	0,6	0,5	0,4	0,3	0,2
8	2,9	2,3	2,1	1,8	1,6	1,5	1,3	1,2	1,1	1,0	0,9	0,8	0,7	0,6	0,5	0,4	0,3
9	3,3	2,6	2,3	2,1	1,8	1,6	1,5	1,4	1,3	1,2	1,1	1,0	0,9	0,8	0,7	0,6	0,5
10	3,7	2,9	2,5	2,2	2,0	1,8	1,7	1,5	1,4	1,3	1,2	1,1	1,0	0,9	0,8	0,7	0,6
11	4,1	3,1	2,8	2,4	2,2	2,0	1,8	1,7	1,5	1,4	1,3	1,2	1,1	1,0	0,9	0,8	0,7
12	4,5	3,4	3,0	2,7	2,4	2,2	2,0	1,8	1,7	1,5	1,4	1,3	1,2	1,1	1,0	0,9	0,8
13	4,9	3,7	3,3	2,9	2,6	2,4	2,2	2,0	1,8	1,7	1,5	1,4	1,3	1,2	1,1	1,0	0,9
14	5,3	4,0	3,5	3,1	2,8	2,5	2,3	2,1	2,0	1,9	1,8	1,6	1,5	1,4	1,3	1,2	1,1
15	5,7	4,3	3,8	3,3	3,0	2,7	2,5	2,3	2,1	2,0	1,9	1,8	1,6	1,5	1,4	1,3	1,2
16	6,1	4,6	4,0	3,6	3,2	2,9	2,7	2,5	2,3	2,1	2,0	1,9	1,8	1,6	1,5	1,4	1,3
17	6,5	4,9	4,3	3,8	3,4	3,1	2,8	2,6	2,4	2,3	2,1	2,0	1,9	1,8	1,6	1,5	1,4
18	6,9	5,1	4,5	4,0	3,6	3,3	3,0	2,8	2,6	2,4	2,3	2,1	2,0	1,9	1,8	1,6	1,5
19	7,3	5,4	4,8	4,2	3,8	3,5	3,2	2,9	2,7	2,5	2,4	2,2	2,1	2,0	1,9	1,8	1,6
20	7,7	5,7	5,0	4,4	4,0	3,6	3,3	3,0	2,8	2,6	2,4	2,3	2,1	2,0	1,9	1,8	1,6

Toe-in/toe-out

Measurement of the rear axle **C5/C6**.

Hang frame gauges, gauge extensions and measuring scales.

Jack up the axle.

Attach wheel adapter and laser projector.

Compensate for the run-out of the wheel with the adjustment knobs on the adapter. Lower the axle.

Calculation

Measure the distance between the measuring scales **A1** and **B1**.

Record results on the Test Report Sheet.

For example: D = 6m.

Wheel **C5** left side

Aim the laser dot at the measuring scale **A1** left front. Read the value.

For example: 110 Make a note on the Test Report Sheet, square **A1**.

Aim the laser dot at measuring scale **B1** left rear. Read the value.

For example: 92 Make a note on the Test Report Sheet, square **B1**.

Rolling direction of wheel **C5** is the difference between the readings of measuring scales **A1** and **B1**. **110 - 92 = +18**.

Divide the difference **+18** by the distance between the measuring scales **A1** and **B1**, **D = 6m**.

Result of **+ 18 : 6 = +3 (mm/m)**.

Write down **3** by **C5/Dm** and circle the (+) sign.

Wheel **C6** right side

Aim the laser dot at the measuring scale **A2** right front. Read the value.

For example: 96 Make a note on the Test Report Sheet, square **A2**

Aim the laser dot at the measuring scale **B2** right rear. Read the value.

For example: 108. Note it down in the Test Report Sheet, square **B2**.

Rolling direction of wheel **C6** is the difference between the measuring scale readings of **A2** and **B2**. **96 - 108 = -12**.

Result of **-12 : 6 = -2 (mm/m)**

Write down **2** by **C6/Dm** and circle the (-) sign.

Result can be calculated by using the table on page 20.

Calculate toe-in/toe-out

C5/Dm = + 3

C6/Dm = - 2

Result = **+ 1 (toe-in)**

If + value is greater, the result is **toe-in**.

If the (-) value is greater, the result is **toe-out**.

Draw rolling direction of the wheel with a solid line.

Draw the position of the axle with a solid line.



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Reg. n° _____

Date _____

Miles/Km _____

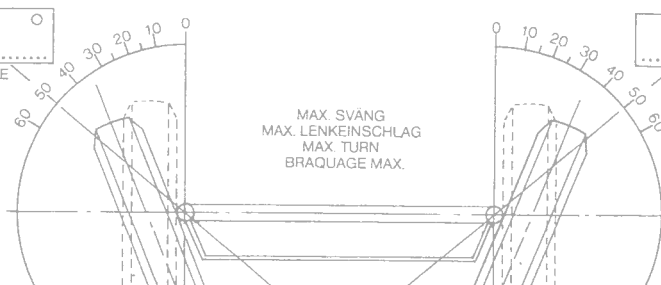
Calculation
Toe-in/toe-out

KURVWINKELDIFFERENS
SPURDIFFERENSWINKEL
TOE-OUT ON TURN
DIFFERENCE DE COURBE D'ANGLE

20° VÄNSTER
LINKS
LEFT
GAUCHE

HÖGER
RECHTS
RIGHT
DROIT

DIFF.



20° HÖGER
RECHTS
RIGHT
DROIT

VÄNSTER
LINKS
LEFT
GAUCHE

DIFF.

A₁ _____
B₁ _____
C₁ $\pm$: D_m =

CAMBER
STURZ
CARROSSAGE
+
-
KPI Spreizung
INCLIN.
PIVOTS
+
-

+
-
CASTER
NACHLAUF
CHASSE

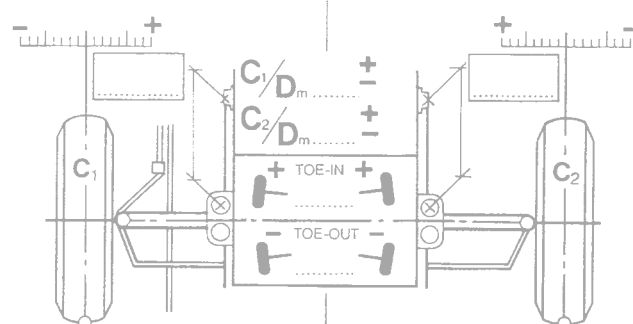
A₁ _____
B₁ _____
C₃ $\pm$: D_m =

CAMBER
STURZ
CARROSSAGE
+
-

A₁ **110**
B₁ **92**
C₅ $\pm$ **18** : D_m **6** =

CAMBER
STURZ
CARROSSAGE
+
-

A₁ 0 100 200 300



A₂ _____
B₂ _____
C₂ $\pm$: D_m =

CAMBER
STURZ
CARROSSAGE
+
-
KPI Spreizung
INCLIN.
PIVOTS
+
-

+
-
CASTER
NACHLAUF
CHASSE

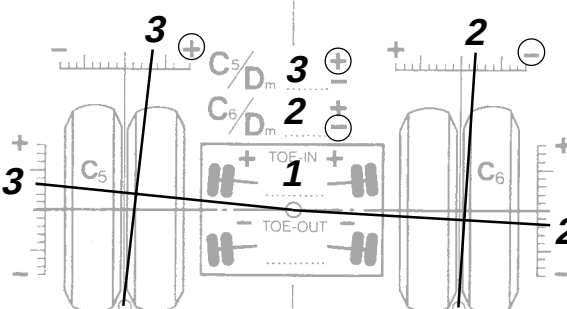
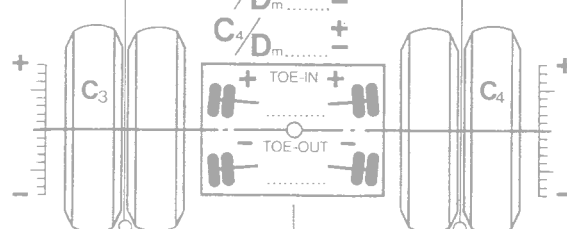
A₂ _____
B₂ _____
C₄ $\pm$: D_m =

CAMBER
STURZ
CARROSSAGE
+
-

A₂ **96**
B₂ **108**
C₆ $\pm$ **12** : D_m **6** =

CAMBER
STURZ
CARROSSAGE
+
-

A₂ 0 100 200 300



B₁ 0 100 200 300

SNEDSTÄLLNING
SCHRAGSTAND
OUT OF SQUARE
ANGLE FAUSSE

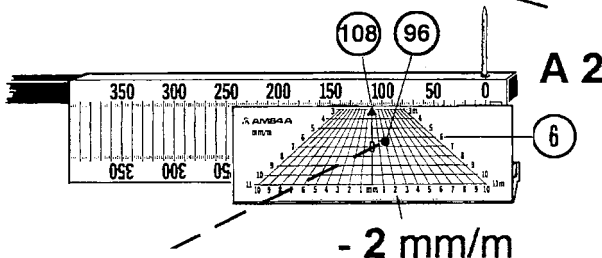
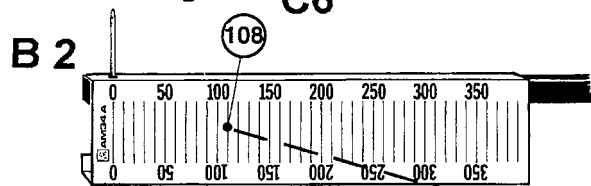
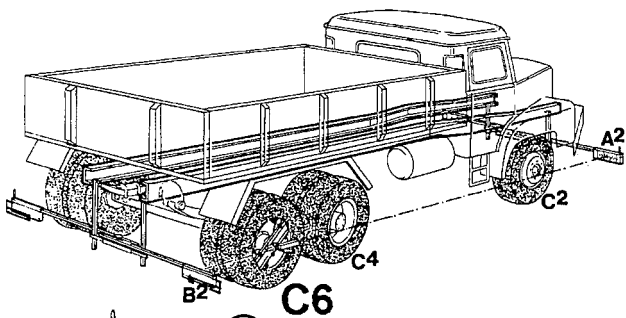
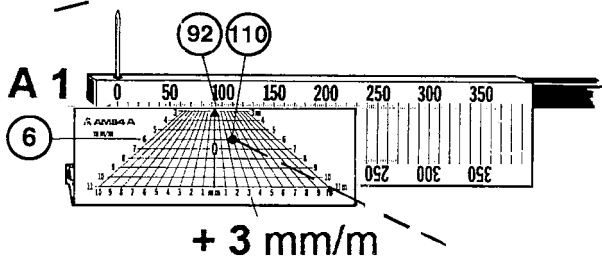
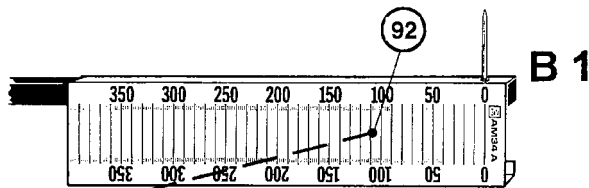
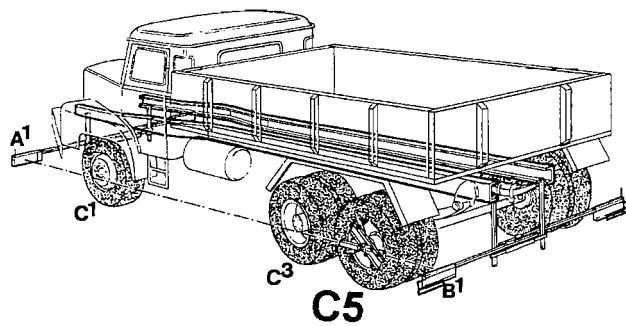
DIFF. C₃/C₄-C₅/C₆
C₄ $\pm$: D_m =
C₆ $\pm$: D_m =

A_{1,2} _____
B_{1,2} _____
= D_m **6** m

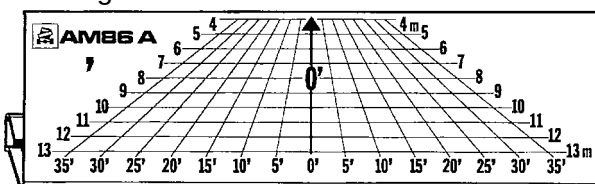


JOSAM laser AM Measuring system

Measurement of rigid axles



Sliding scale AM86 A



Toe-in/toe-out

Measuring with sliding scale AM84 A axle C5/C6.

Hang frame gauges, gauge extensions and measuring scales.

Jack up the axle.

Attach wheel adapter and laser projector.

Compensate for run-out of the wheel using the wheel adapter adjustment knobs. Lower the axle. Place the sliding scales onto both front measuring scales A1 and A2.

Measure the distance between measuring scales A1 and B1 and write it on the Test Report Sheet.

For example: D = 6m.

Wheel C5 left side

Aim the laser dot at the measuring scale B1 left rear.

Read the value. **For example: 92**

Move the sliding scale on the measuring scale A1 left front and set the 0-line on 92.

Aim the laser dot on the 6m distance line on the sliding scale.

On the bottom row read off rolling direction of wheel **+3 mm/m**.

Write down **3** at C5/Dm in the middle field and circle the (+) sign.

Wheel C6 right side

Aim laser dot at the measuring scale B2 right rear.

Read the value. **For example: 108.**

Move the sliding scale on the measuring scale A2 right front and set the 0-line on 108.

Aim the laser dot at the horizontal 6m distance line on the sliding scale.

On the bottom row read off rolling direction of wheel **-2 mm/m**.

Write down **2** at C6/Dm in the middle field and circle the (-) sign.

Draw in rolling direction of wheel with a solid line. Draw the axle position with a solid line.

If the laser dot lies **inside the 0-line** on the sliding scale it is a **(+) value**.

If the laser dot lies **outside the 0-line** on the sliding scale it indicates a **(-) value**.

Calculate toe-in and toe-out

$$C5/Dm = +3$$

$$C6/Dm = -2$$

$$\text{Result} = +1(\text{toe-in})$$

Sliding Scale AM86 A

On the sliding scale AM86 A the result on the bottom row is indicated in minutes instead of mm/m.



JOSAM laser AM Measuring system

Measurement of rigid axles



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Reg. n° _____

Date _____

Miles/Km _____ Sign _____

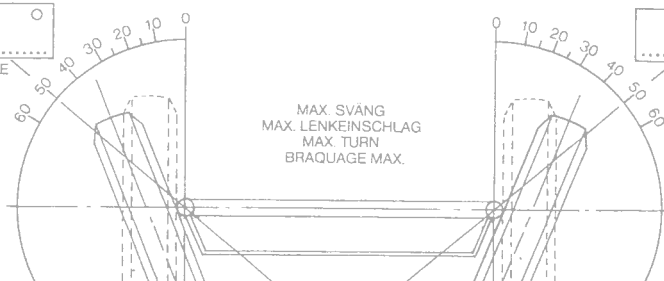
**With sliding scale AM84 A
toe-in/toe-out**

KURVINKELDIFFERENS
SPURDIFFERENSWINKEL
TOE-OUT ON TURN
DIFFERENCE DE COURBE D'ANGLE

20°
VÄNSTER
LINKS
LEFT
GAUCHE

HÖGER
RECHTS
RIGHT
DROIT

DIFF.



20°
HÖGER
RECHTS
RIGHT
DROIT

VÄNSTER
LINKS
LEFT
GAUCHE

DIFF.

A₁
B₁
C₁ $\pm$: **D_m** =

CAMBER
STURZ
CARROSSAGE
+

+
CASTER
NACHLAUF
CHASSE

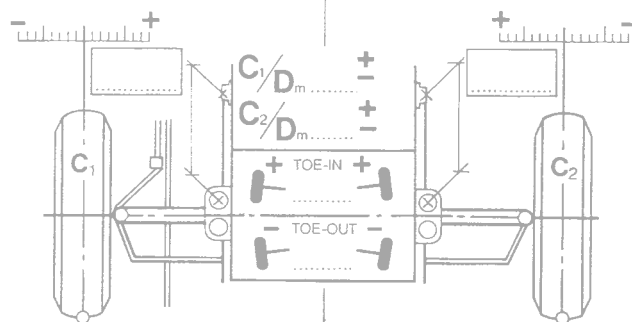
A₁
B₁
C₃ $\pm$: **D_m** =

CAMBER
STURZ
CARROSSAGE
+

A₁ **110**
B₁ **92**
C₅ $\pm$ **18** : **D_m** **6** =

CAMBER
STURZ
CARROSSAGE
+

A₁ **A₂**



A₂
B₂
C₂ $\pm$: **D_m** =

CAMBER
STURZ
CARROSSAGE
+

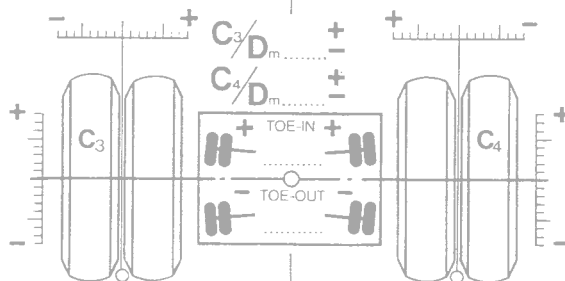
+
CASTER
NACHLAUF
CHASSE

A₂
B₂
C₄ $\pm$: **D_m** =

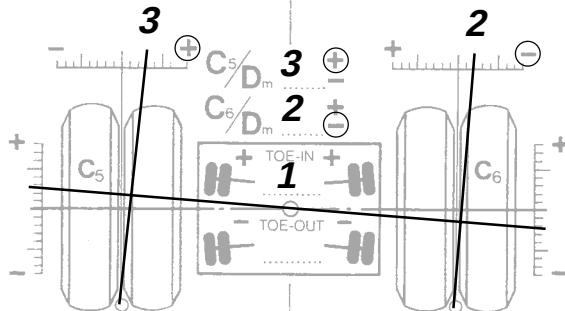
VÄNSTER
LINKS
LEFT
GAUCHE
+

CAMBER
STURZ
CARROSSAGE
+

A₁ **A₂**



A₁ **A₂**

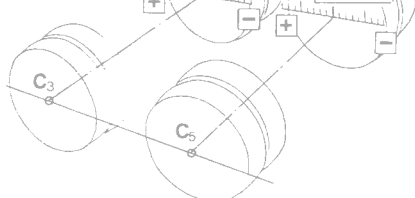


A₁ **A₂**

SNEDSTÄLLNING
SCHRAGSTAND
OUT OF SQUARE
ANGLE FAUSSE

DIFF. C₃/C₄ - C₅/C₆

C₄ $\pm$ **C₆** $\pm$ =



A_{1,2}
B_{1,2}
D_m **6**

AM39 1+2+4

T 27 1-2-3-4 9405



Out of square axle

Calculation of out of square axle

Toe-in has been calculated to +1 mm/m.

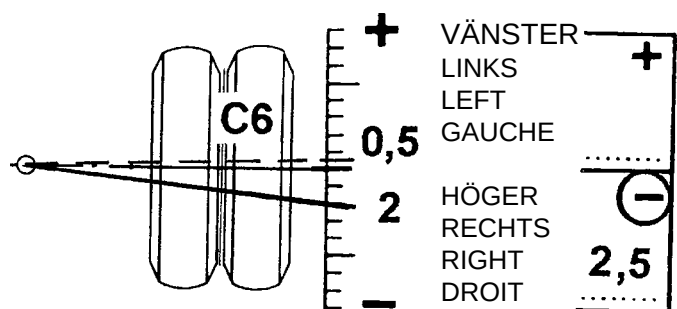
$$C5/Dm = +3 \text{ (toe-in)}$$

$$C6/Dm = -2 \text{ (toe-out)}$$

$$\text{Result} = +1 \text{ (toe-in)}$$

Toe-in +1 mm/m is divided by 2 = 0.5 mm/m.

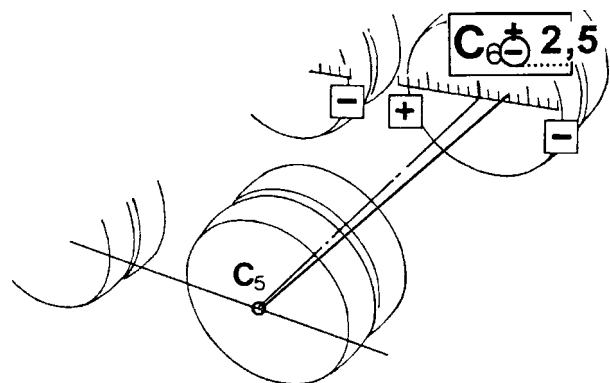
Draw a dotted line on the Test Report Sheet from the center, through wheel C6 to +0.5 on the scale to the right. Write down the value 0.5 on the dotted line. Put in the number 2 on the solid drawn line.



The out of square is $0.5 + 2 = 2.5 \text{ mm/m}$.

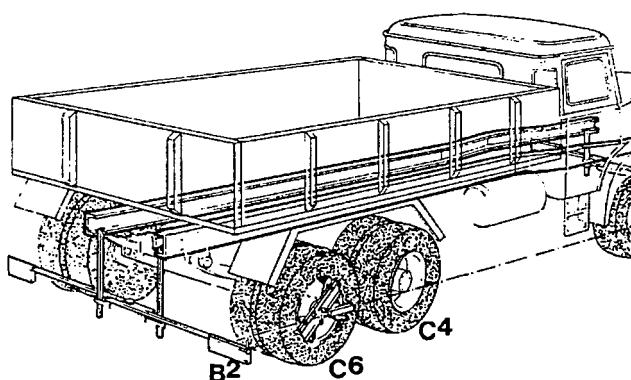
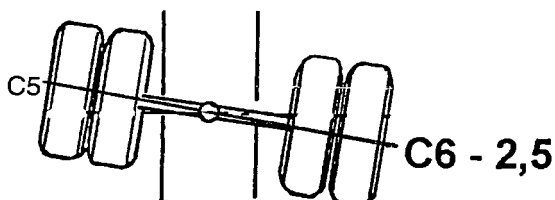
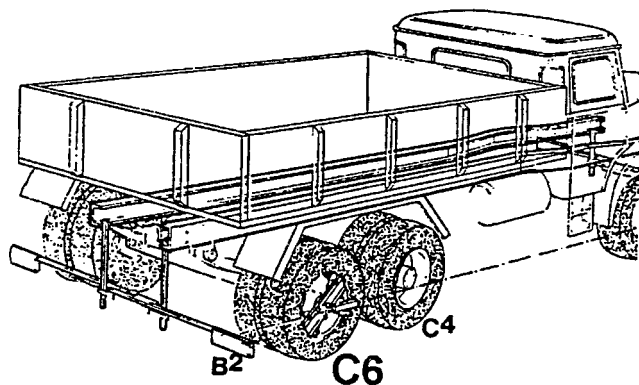
Write down 2.5 and circle a (-) sign in the square "right".

Write down the out of square reading -2.5 at C6 on the lower picture and draw it with a solid line C5/C6.



If there is an out of square position to the **right**, note the value as a minus (-).

If the out of square position is to the **left**, note the value as a plus (+).





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Reg. n° _____
Date _____
Miles/Km _____

Calculation
Out of square axle

KURVINKELDIFFERENS
SPURDIFFERENSWINKEL
TOE-OUT ON TURN
DIFFERENCE DE COURBE D'ANGLE

20° VÄNSTER
LINKS
LEFT
GAUCHE

HÖGER
RECHTS
RIGHT
DROIT

DIFF.

MAX. SVÄNG
MAX. LENKEINSCHLAG
MAX. TURN
BRAQUAGE MAX.

20° HÖGER
RECHTS
RIGHT
DROIT

VÄNSTER
LINKS
LEFT
GAUCHE

DIFF.

A₁ 0 100 200 300 A₂ 300 200 100 0

B₁ B₂

C₁ : D_m = C₂ : D_m =

CAMBER STURZ CARROSSAGE + - KPI Spreizung INCLIN. PIVOTS + -

CASTER NACHLAUF CHASSE + -

A₁ A₂

B₁ B₂

C₃ : D_m = C₄ : D_m =

CAMBER STURZ CARROSSAGE + -

VÄNSTER LINKS LEFT GAUCHE + -

HÖGER RECHTS RIGHT DROIT + -

CAMBER STURZ CARROSSAGE + -

A₁ 110 A₂ 96

B₁ 92 B₂ 108

C₅ : D_m = 18 : 6 = C₆ : D_m = 12 : 6 =

CAMBER STURZ CARROSSAGE + -

VÄNSTER LINKS LEFT GAUCHE + -

HÖGER RECHTS RIGHT DROIT + -

CAMBER STURZ CARROSSAGE + -

3 2 0,5 2 2,5

B₁ 0 100 200 300 B₂ 300 200 100 0

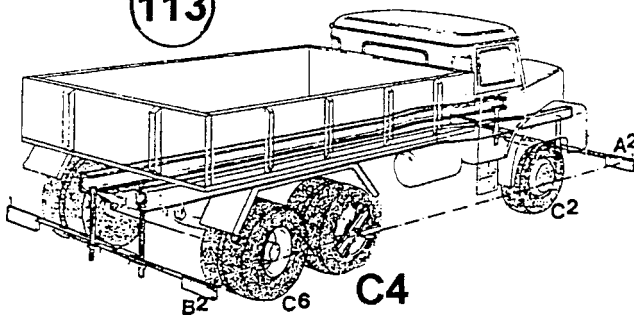
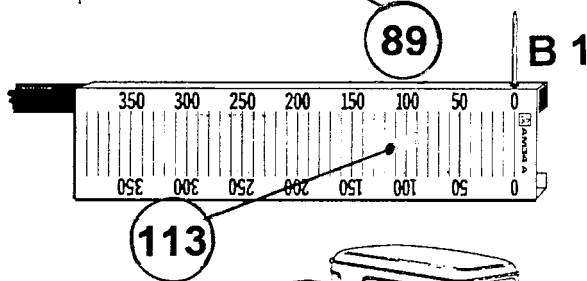
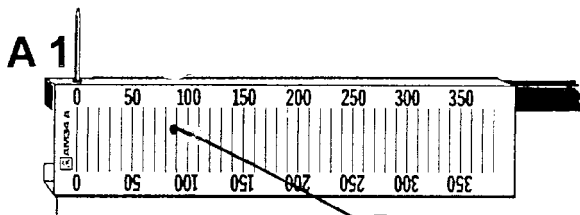
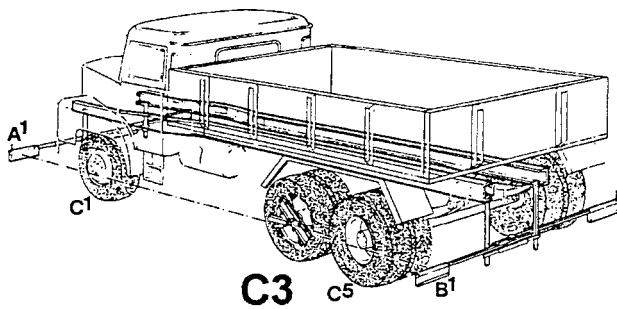
SNEDSTÄLLNING
SCHRAGSTAND
OUT OF SQUARE
ANGLE FAUSSE

DIFF. C₃/C₄ - C₅/C₆ C₄ - C₆ = 2,5

A_{1,2} = D 6 m
B_{1,2}

AM39 1+2+4

T 27 1-2-3-4 9405



Total difference between front and rear measuring scales
Distance in meters between the measuring scales

C/D	mm/m									
	3	3,5	4	4,5	5	5,5	6	6,5	7	7,5
3	1	0,9	0,8	0,7	0,6	0,5	0,4	0,3	0,2	0,1
4	1,3	1,1	1	0,9	0,8	0,7	0,6	0,5	0,4	0,3
5	1,7	1,4	1,3	1,1	1	0,9	0,8	0,7	0,6	0,5
6	2	1,7	1,5	1,3	1,2	1,1	1	0,9	0,8	0,7
7	2,3	2	1,8	1,6	1,4	1,3	1,2	1,1	1	0,9
8	2,7	2,3	2	1,8	1,6	1,5	1,3	1,2	1,1	1
9	3	2,6	2,3	2	1,8	1,6	1,5	1,4	1,3	1,2
10	3,3	2,9	2,5	2,2	2	1,8	1,7	1,5	1,4	1,3
11	3,7	3,1	2,8	2,4	2,2	2	1,8	1,7	1,6	1,5
12	4	3,4	3	2,7	2,4	2,2	2	1,9	1,8	1,7
13	4,3	3,7	3,3	2,9	2,6	2,4	2,2	2	1,9	1,8
14	4,7	4	3,5	3,1	2,8	2,5	2,3	2,1	2	1,9
15	5	4,3	3,8	3,3	3	2,7	2,5	2,3	2,1	2
16	5,3	4,6	4	3,6	3,2	2,9	2,7	2,5	2,3	2,1
17	5,7	4,9	4,3	3,8	3,4	3,1	2,8	2,6	2,4	2,2
18	6	5,1	4,5	4	3,6	3,3	3	2,8	2,6	2,4
19	6,3	5,4	4,8	4,2	3,8	3,5	3,2	2,9	2,7	2,5
20	6,7	5,7	5	4,4	4	3,6	3,3	3	2,9	2,7

Toe-in/toe-out

Measuring of axle **C3/C4**.

Preparations are the same as the measurement of rear axle **C5/C6**.

Calculation

Measure the distance between the measuring scales A1 and B1. **For example: D = 6m.**

Wheel **C3** left side

Aim laser dot at the measuring scale **A1** left front.
Read the value.

For example: 89 Write down the result on the Test Report Sheet, square **A1**.

Aim the laser dot at the measuring scale **B1** left rear.
Read the value.

For example: 113 Record this on the Test Report Sheet, square **B1**.

Rolling direction of wheel C3 is the difference between the readings of measuring scales A1 and B1. **89 - 113 = -24**

Divide the difference **-24** by the distance between the measuring scales A1 and B1, **D = 6m.**

The result: **-24 : 6 = -4 (mm/m).**

Write down **4** at **C3/Dm** and circle a **(-)** sign.

Wheel **C4** right side

Aim the laser dot at the measuring scale **A2** right front.
Read the value.

For example: 108 Write down the result on the Test Report Sheet, square **A2**.

Aim the laser dot at the measuring scale **B2** right rear.
Read the value.

For example: 96 Write down the result in the Test Report Sheet, square **B2**.

Rolling direction of wheel C4 is the difference between the readings of measuring scales A2 and B2. **108 - 96 = +12.**

The result: **+12 : 6 = +2 (mm/m).**

Write down **2** at **C4/Dm** and circle a **(+)** sign.

The result can be calculated by using the table to the left.

Calculate toe-in/toe-out

C3/Dm = - 4

C4/Dm = +2

Result = **- 2 (toe-out)**

If the **(-)** value is greater, the result is **toe-out**.

If the **(+)** value is greater, the result is **toe-in**.

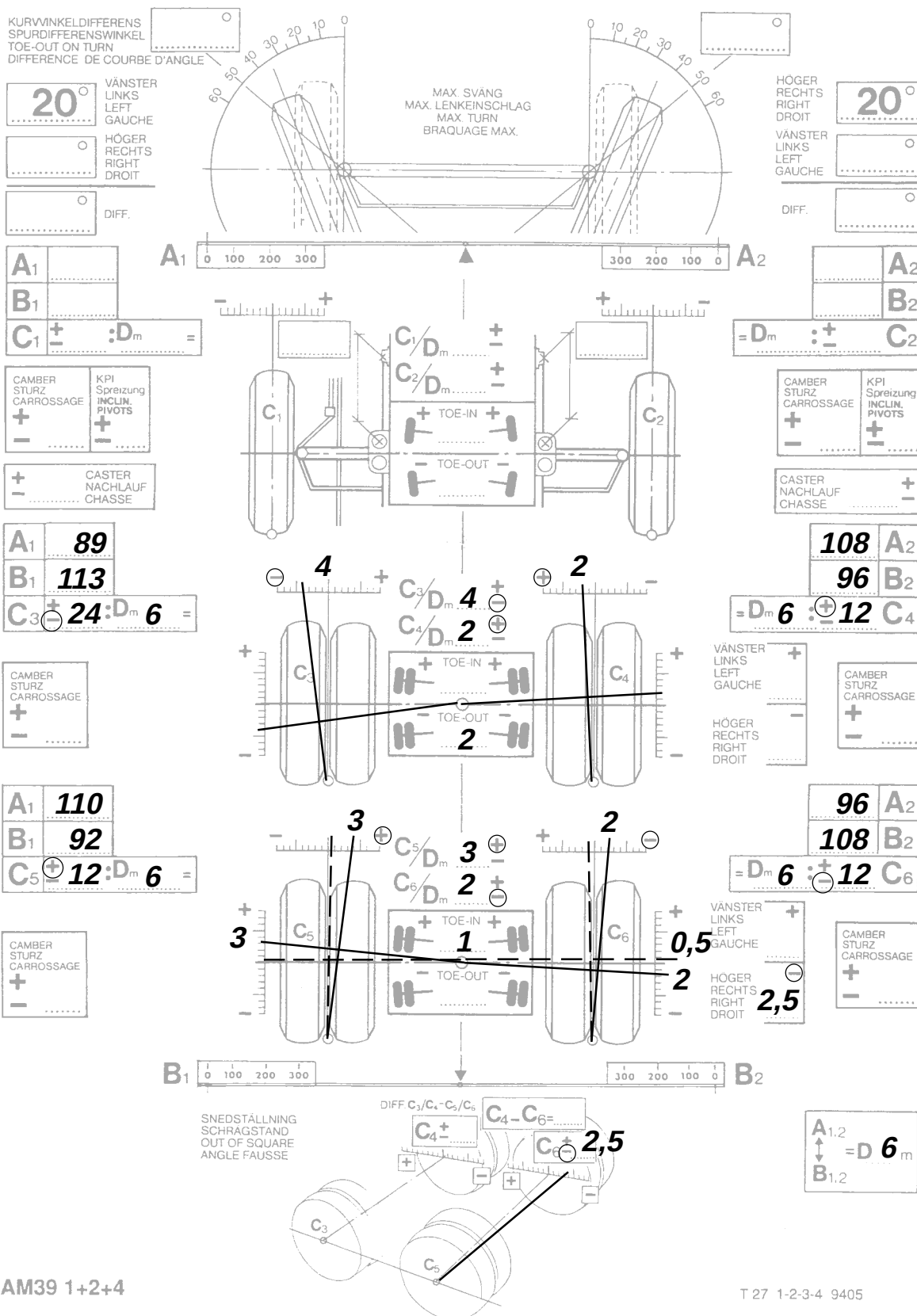
Draw rolling direction of the wheel with a solid line.
Draw the axle location with a solid line.



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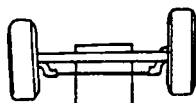
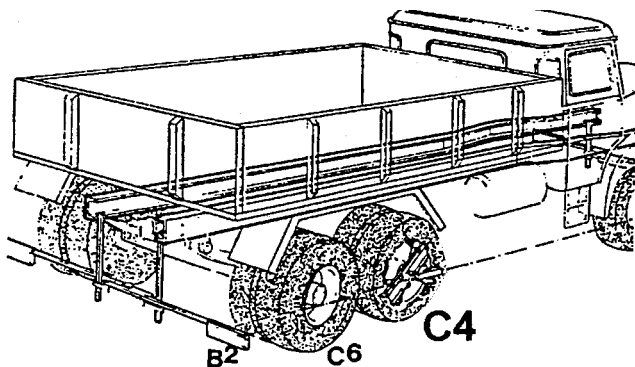
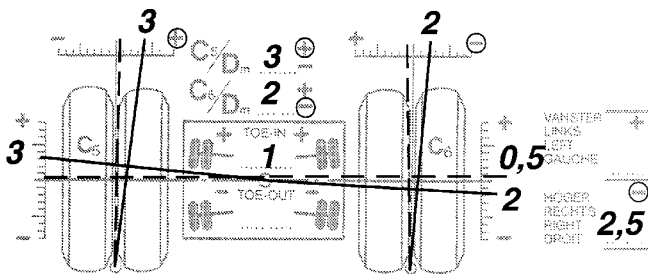
Reg. n° _____
Date _____
Miles/Km _____

Calculation
Toe-in/toe-out

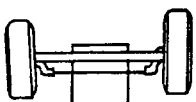
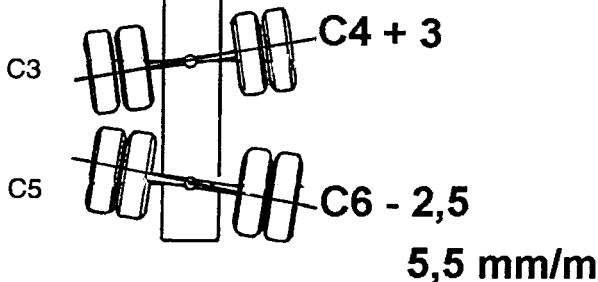




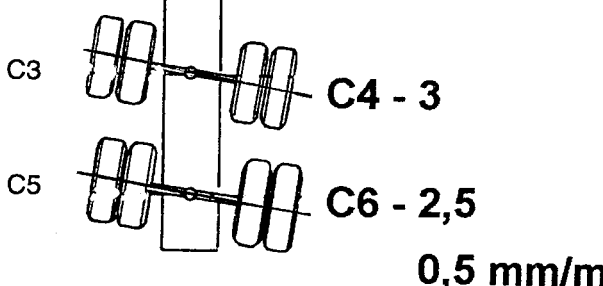
Out of square axle



The axles are not parallel.



The axles are parallel.



According to the Test Report Sheet the axle C5/C6 has an out of square measurement of **-2.5** (see Test Report Sheet pg. 31).

Measuring of out of square axle C3/C4
Toe-out has been calculated at **-2 mm/m**.

C3/Dm = **-4** (toe-out)

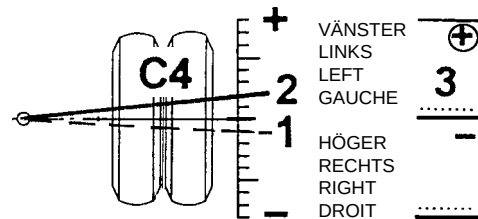
C4/Dm = **+2** (toe-in)

Result **-2** (toe-out)

Toe-out **-2 mm/m** divided by **2** = **-1 mm/m**.

Draw a dotted line on the Test Report Sheet from the center for wheel C3 and C4 to **-1** on the scale at the side of the wheel.

Write down **-1** on the dotted line on the right scale
Write down **2** on the solid line.



Out of square is **1 + 2 = 3 mm/m**.

Write down **3** and circle **(+)** sign in the "left" square.

Write down out of square **+3** at C4 on the lower picture and draw it with a solid line C3/C4.

Axle C3/C4 is out of square **+3** to the left.

Axle C5/C6 is out of square **-2.5** to the right.

The difference between the axles is evident at wheels C4 & C6. **+3 is added to -2.5 = 5.5 mm/m**.
When one axle is out of square to the left and the other to the right, the two values should be added together. **The axles are not parallel.**

When the axles are out of square in the same direction, the smaller value should be subtracted from the larger value.

For example:

Axle C3/C4 is out of square **-3** to the right.

Axle C5/C6 is out of square **-2.5** to the right.

-3 - -2.5 = 0.5 mm/m.

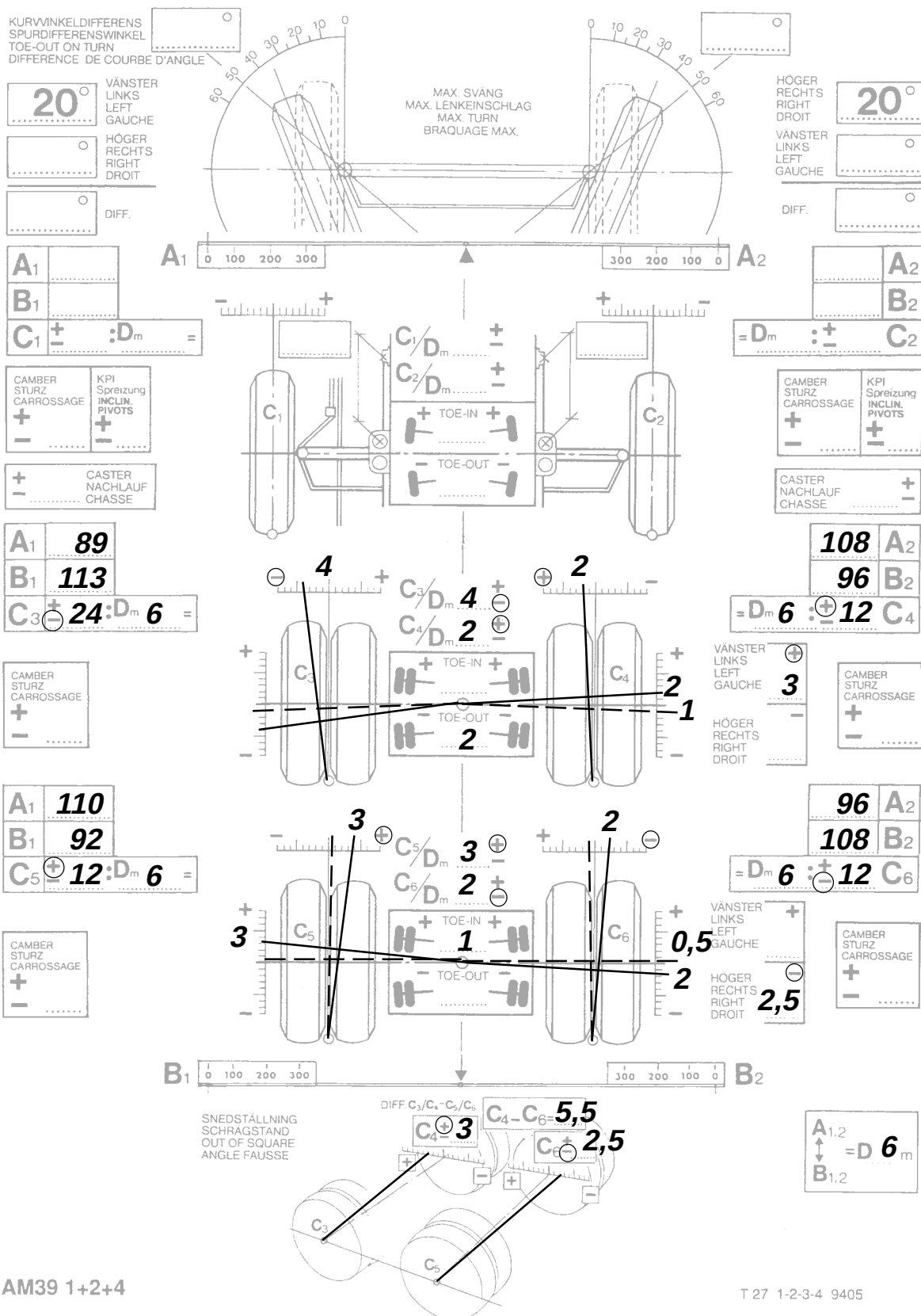
The axles are parallel.



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Reg. n° _____
Date _____
Miles/Km _____

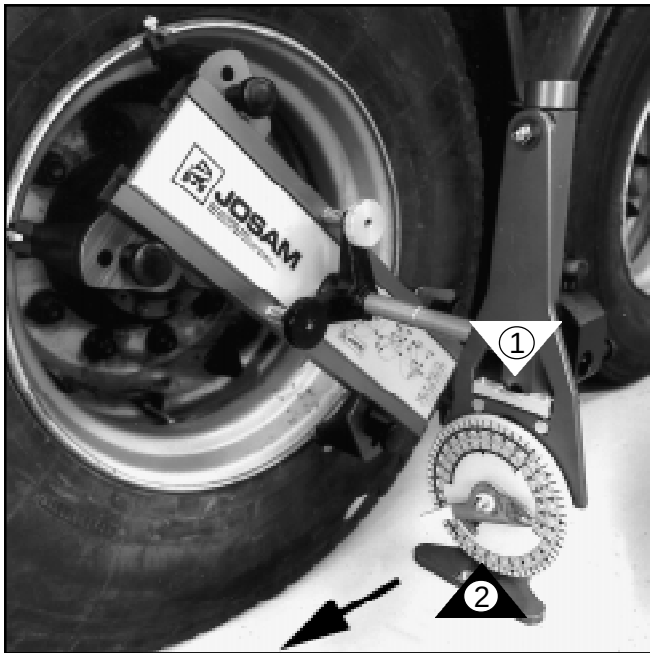
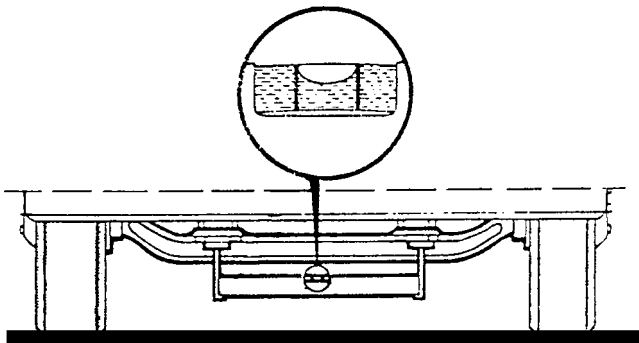
Calculation
Out of square axle



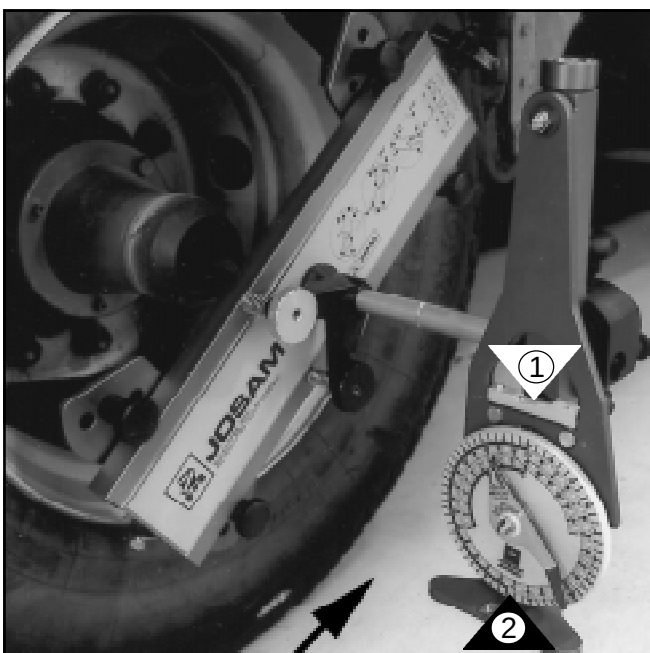


JOSAM laser AM Measuring system

Measurement of rigid axles



Left



Right

Measurement of camber

For camber measurements the axle must be in a horizontal position. The wheel should be on the ground and "loaded".

Hang the frame gauges, gauge extensions and measuring scales. Jack up the axle. Attach the wheel adapter and laser projector. Run-out of the wheel can be compensated with the adjustment knobs on the adapter.

Lower the axle to the floor.

If the floor is not flat (level) or if the size of the wheels is different, check the position of the axle with a spirit level.

The spirit level can be hung under the axle with a magnetic hanger, for example, AM98.

Look for a place on the axle that is machined.

Do not measure on the underside of the springs if there are different size spring spacers between the axle and the springs.

Adjust unevenness by placing plates under one wheel until the axle is level.

If the floor is flat (level) check the measurement between the floor and the hub, bead of rim, or something similar with the axle lowered and the wheel on the floor.

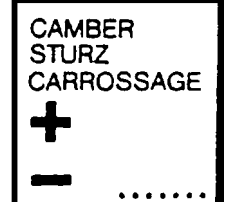
Attach wheel alignment gauge AM300/301 onto the spindle of the wheel adapter through the hole for camber and caster measurements.

Position the wheel alignment gauge vertically using the top bubble and lock into place with the lower knob.

1 Turn the measuring scale so that the measuring bubble shows a horizontal position.

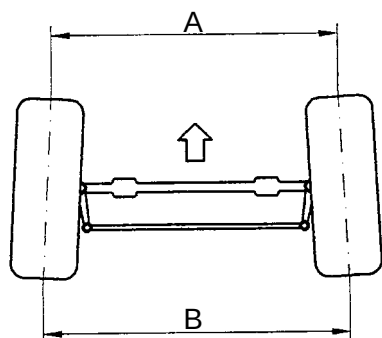
2 Read the camber value on the outer scale with the arrow on the foot plate.

Write down the value on the Test Report Sheet. Circle whether it is a (+) or a (-) value.



NOTE!

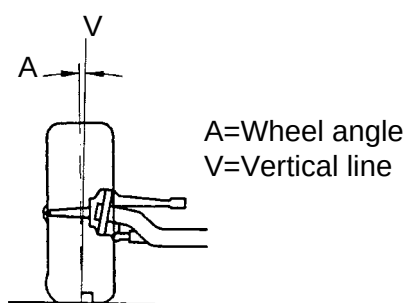
When taking camber measurements the axle must be in a horizontal position. The wheel should be on the floor "under load".



Toe-in/toe-out

Toe-in (+): distance between the front wheels (A) is less than distance between the back wheels (B).

Toe-Out (-): distance between the back wheels (B) is less than distance between the front wheels (A).
Measurement in mm/m or minutes ´.



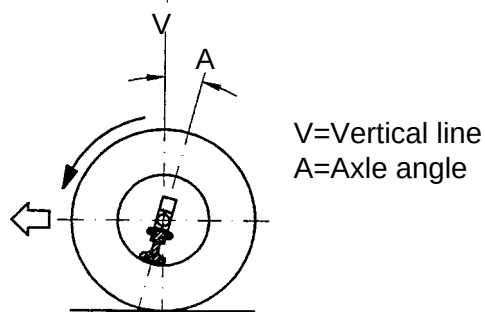
Camber

Leaning of the wheel outward or inward.

The upper part of the wheel leans outward: positive (+) camber.

The upper part of the wheel leans inward: negative (-) camber.

Measurement in degrees ° and minutes ´.



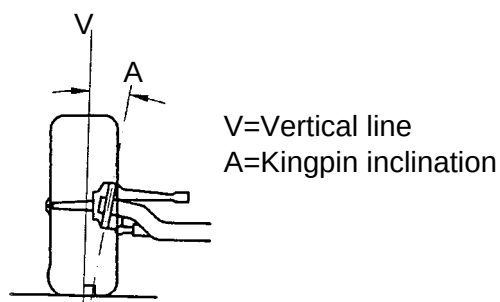
Caster

The backward or forward inclination of the kingpin, as seen from the side.

The upper part of the kingpin inclines backward: positive (+) caster.

The upper part of the kingpin inclines forward: negative (-) caster.

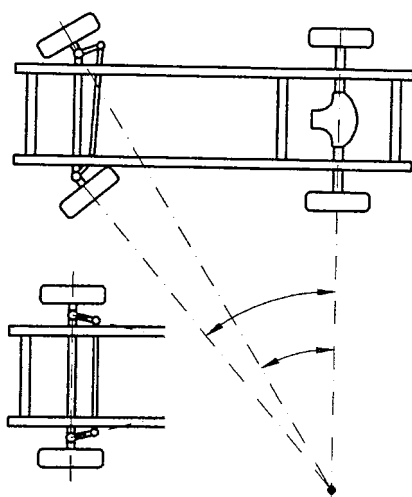
Measurement in degrees ° and minutes ´.



KPI (kingpin inclination)

The upper part of the kingpin inclines inward when seen from the front or the back.

Measurement in degrees ° and minutes ´.

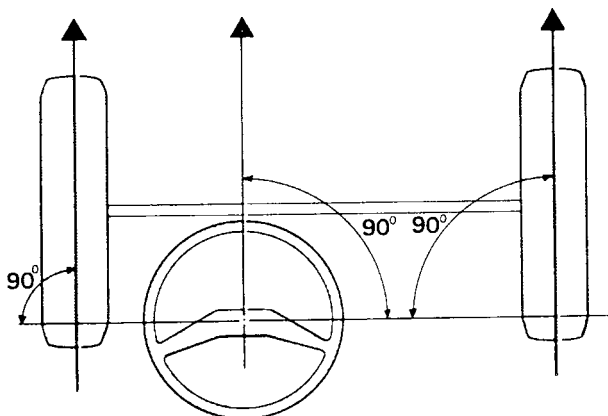
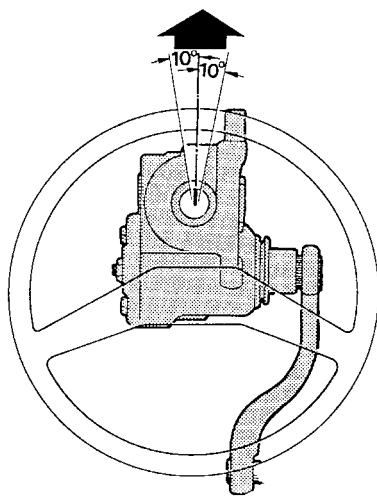
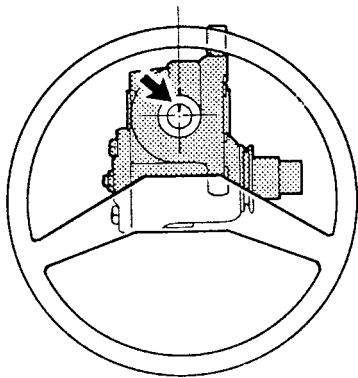


toe out on turn (TOOT)

Different turn for outer and inner wheels during turning.

The angle on the steering arms in relation to the wheel makes the inner wheel rotate more than the outer wheel.

Measurement in degrees ° and minutes ´.



Checking the center point of the steering box when it is marked.

Hang the frame gauges, gauge extensions and measuring scales.

Jack up the axle.

Attach the wheel adapter and laser projector.

Run-out of the wheel can be compensated with the adjustment knobs on the adapter.

Manufacturers mark the center point on the steering box and the input shaft, as well as the maximum deviation between the mark on the input axle and the mark on the steering box.

Maximum deviation is normally $\pm 10^\circ$.

Turn the steering wheel until the mark on the steering box is in the designated center point. Aim the laser projector from the wheel that is on the same side as the steering box is located, toward the front measuring scale **A** and read the value. Rotate the laser projector and aim it at the rear measuring scale **B** and read the value.

If the values are the same on both measuring scales, the center point of the steering box is in agreement with the straight ahead position of the wheels.

If the values on the measuring scales **A** and **B** do not agree, check the deviation.

The normal deviation between the middle position of the steering box and the straight ahead position is **1° per meter**, **1° = 17.5 mm per m**.

The difference between the values on the front and rear measuring scales is given in millimeters (mm). The distance between the measuring scales is in meters (m).

The value in (mm) divided by the distance between the measuring scale in (m), gives mm/m.

For example:

The front measuring scale A = 115

Rear measuring scale B = 220

The distance between the measuring scales A and B = 6m

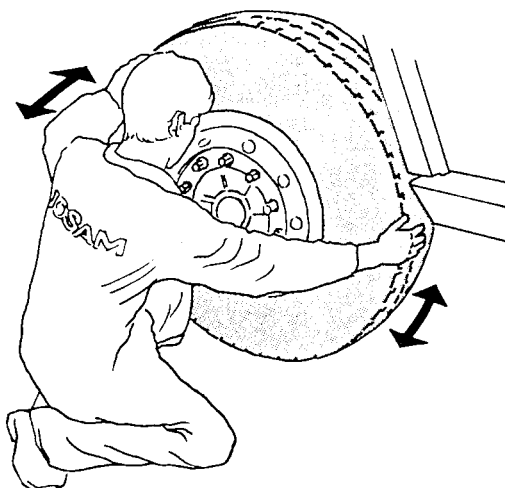
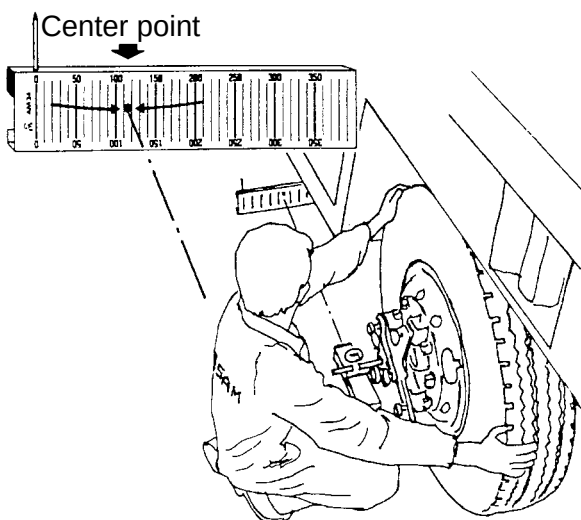
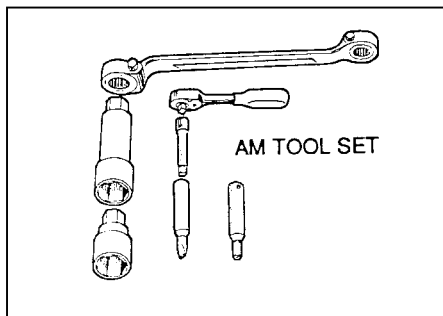
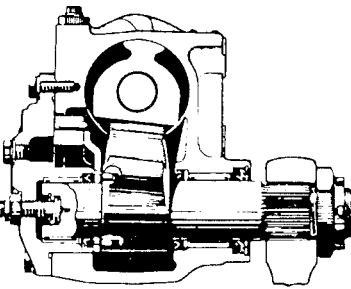
$220 - 115 = 105 : 6 = 17.5$

The deviation is 17.5 mm/m, or 1° per meter.

If the deviation is more than **1°/m**, the length of the steering linkage should be adjusted so that the center point on the steering box corresponds with the straight ahead position of the wheel.



Adjusting
screw and
lock nut



Checking the center point of the steering box when it is not marked.

On a steering box with an adjusting screw and lock nut, one can look for the center point.

Hang the frame gauges, gauge extensions and measuring scales.

Jack up the axle.

Attach the wheel adapter and laser projector.

Compensate for run-out of the wheel with the adjustment knobs on the adapter.

Loosen the lock nut and tighten the adjustment screw until resistance is felt when the wheel is turned by hand back and forth over the center point. Tighten the lock nut.

This adjustment is easier to perform with the AM TOOL SET.

Aim the laser projector at the front measuring scale. Turn the wheel by hand from the **inside and out toward the middle** until light resistance is felt. Read the value on the measuring scale. **For example: 130.** Turn the wheel then from the **outside and in toward the middle** until light resistance is felt. Read the value on the measuring scale. **For example: 100.**

The center point is 115, which is halfway between 130 and 100.

Position the wheel with the laser dot on the front measuring scale **A** at 115.

Aim the laser projector at the rear measuring scale **B**. If the values are the same on both measuring scales, the center point of the steering box corresponds with the straight ahead position of the wheel.

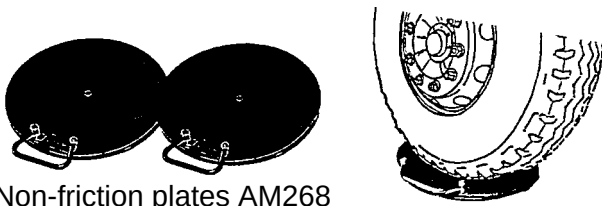
If the values on the measuring scales **A** and **B** do not agree, check the deviation.

The normal deviation between the center point of the steering box and the straight ahead position of the wheel is **1° per meter, 1° = 17.5 mm per m.**

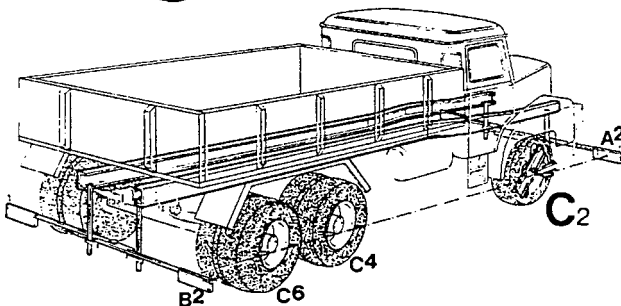
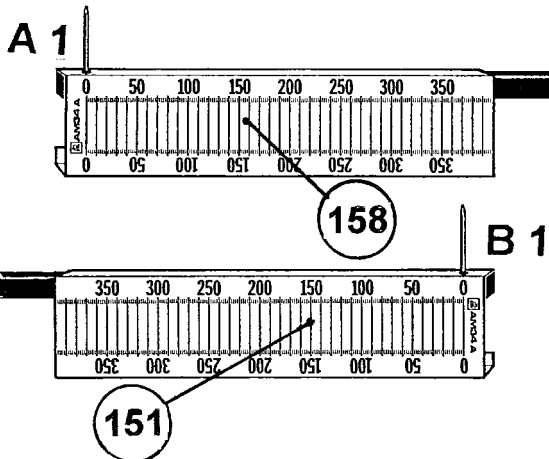
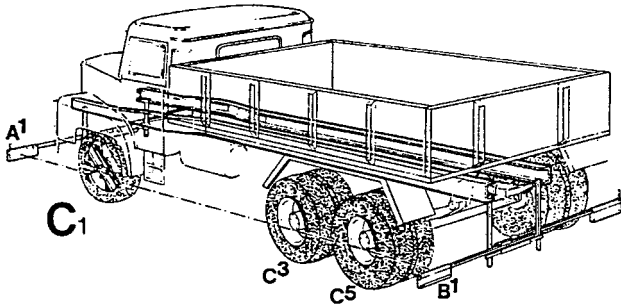
If the deviation is more than **1°/m**, the length of the steering linkage should be adjusted so that the center point on the steering box corresponds with the straight ahead position of the wheel.

Adjust the play in the steering box with the adjusting screw until no resistance is felt when the wheel is turned by hand over the center point.

The wheel must be possible to turn without feeling resistance.



Non-friction plates AM268



D _m \ C _{mm}		C/D mm m																
C _{mm}	D _m	3	3,5	4	4,5	5	5,5	6	6,5	7	7,5	8	8,5	9				
		3	3,5	4	4,5	5	5,5	6	6,5	7	7,5	8	8,5	9				
3	3	1	0,9	0,8	0,7	0,6		0,5		0,4								
4	4	1,3	1,1	1	0,9	0,8		0,7		0,6		0,5		0,4				
5	5	1,7	1,4	1,3	1,1	1	0,9	0,8		0,7		0,6		0,5				
6	6	2	1,7	1,5	1,3	1,2	1,1	1		0,9		0,8		0,7				
7	7	2,3	2	1,8	1,6	1,4	1,3	1,2	1,1	1		0,9		0,8				
8	8	2,7	2,3	2	1,8	1,6	1,5	1,3	1,2	1,1		1		0,9				
9	9	3	2,6	2,3	2	1,8	1,6	1,5	1,4	1,3	1,2	1,1		1				
10	10	3,3	2,9	2,5	2,2	2	1,8	1,7	1,5	1,4	1,3	1,2	1,1	1				
11	11	3,7	3,1	2,8	2,4	2,2	2	1,8	1,7	1,6	1,5	1,4	1,3	1,2	1,1			
12	12	4	3,4	3	2,7	2,4	2,2	2	1,8	1,7	1,6	1,5	1,4	1,3	1,2	1,1		
13	13	4,3	3,7	3,3	2,9	2,6	2,4	2,2	2	1,9	1,7	1,6	1,5	1,4	1,3	1,2	1,1	
14	14	4,7	4	3,5	3,1	2,8	2,5	2,3	2,1	2	1,9	1,8	1,6	1,5	1,4	1,3	1,2	1,1
15	15	5	4,3	3,8	3,3	3	2,7	2,5	2,3	2,1	2	1,9	1,8	1,7	1,6	1,5	1,4	1,3
16	16	5,3	4,6	4	3,6	3,2	2,9	2,7	2,5	2,3	2,1	2	1,9	1,8	1,7	1,6	1,5	1,4
17	17	5,7				3,4	3,1	2,8	2,6	2,4	2,2	2	1,9	1,8	1,7	1,6	1,5	1,4

Toe-in/toe-out

Hang the frame gauges, gauge extensions and measuring scales. Jack up the front axle. Attach the wheel adapter and laser projector. Compensate for run-out of the wheel with the adjustment knobs on the adapter.

Position the left or the right wheel in a straight ahead position.

Same value on measuring scales A and B. Lower the wheels onto the floor.

In order to make it easier to adjust toe-in/toe-out, it is an advantage to place non-friction plates AM268 under the two wheels.

Measure the distance between measuring scales A - B. Write down the result on the Test Report Sheet.

For example: D = 7m.

Wheel C1 left side

Aim the laser dot at the measuring scale **A1** left front. Read the value. **For example: 158** Write it down on the Test Report Sheet in **square A1**. Aim the laser dot at measuring scale **B1** left rear. Read the value. **For example: 151** Write down the result in the **B1 square** on the Test Report Sheet.

Rolling direction of wheel C1

The difference between the measuring scales **A1** and **B1**: **158 - 151 = +7**.

Divide **+7** by the distance between the measuring scale: **D = 7m**. **+7 : 7 = +1 (mm/m)**.

Write down **1** in the middle of the Test Report Sheet at **C1/Dm** and circle a **(+)** sign.

Draw the rolling direction of wheel with a solid line.

Wheel C2 right side

Aim the laser dot at the measuring scale **A2** right front. Read the value. **For example: 140** Write it down on the Test Report Sheet in **square A2**.

Aim the laser dot at the measuring scale **B2** right rear. Read the value. **For example: 154** Write the result on the Test Report Sheet in **square B2**.

Rolling direction of wheel C2

The difference between measuring scales **A2** and **B2**: **140 - 154 = -14**. Divide **-14** by the distance between the measuring scales **D = 7m**.

-14 : 7 = -2 (mm/m).

Write down **2** in the middle of the Test Report Sheet at **C2/Dm** and circle a **(-)** sign. Draw the rolling direction of the wheel with a solid line.

Calculating toe-in/toe-out

Wheel **C1/Dm** **+ 1 mm/m**

Wheel **C2/Dm** **- 2 mm/m**

Result **- 1 mm/m toe-out**

If **(+)** value is greater, the result is **toe-in**.

If **(-)** value is greater, the result is **toe-out**.

The result can be calculated by using the table to the left.



Reg. n° _____
Date _____
Miles/Km _____

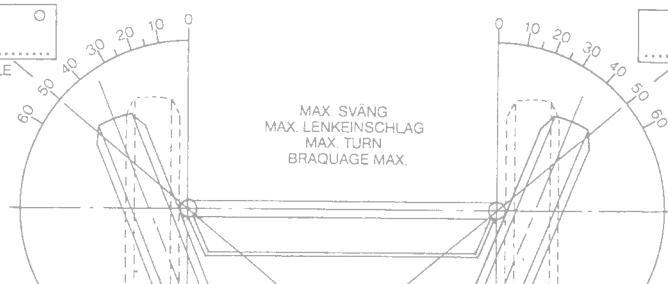
Calculations
Toe-in/toe-out

KURVWINKELDIFFERENS
SPURDIFFERENSWINKEL
TOE-OUT ON TURN
DIFFERENCE DE COURBE D'ANGLE

20° VÄNSTER
LINKS
LEFT
GAUCHE

0° HÖGER
RECHTS
RIGHT
DROIT

0° DIFF.



20° HÖGER
RECHTS
RIGHT
DROIT

0° VÄNSTER
LINKS
LEFT
GAUCHE

0° DIFF.

A₁ 158
B₁ 151
C₁ 7 : D_m 7 =

CAMBER
STURZ
CARROSSAGE
+

KPI
Spreizung
INCLIN.
PIVOTS
+

+ CASTER
NACHLAUF
CHASSE

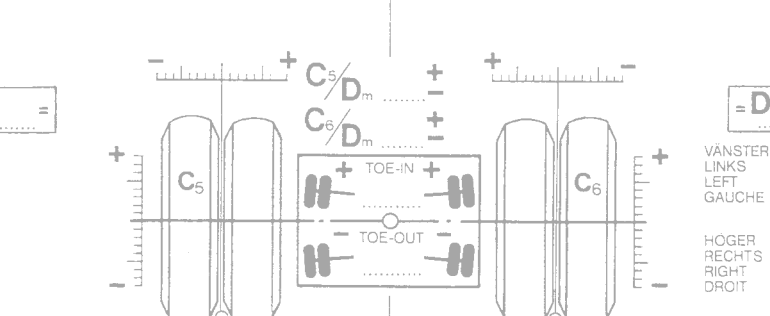
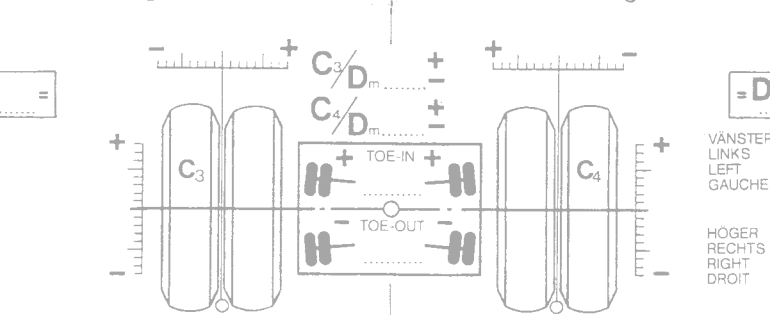
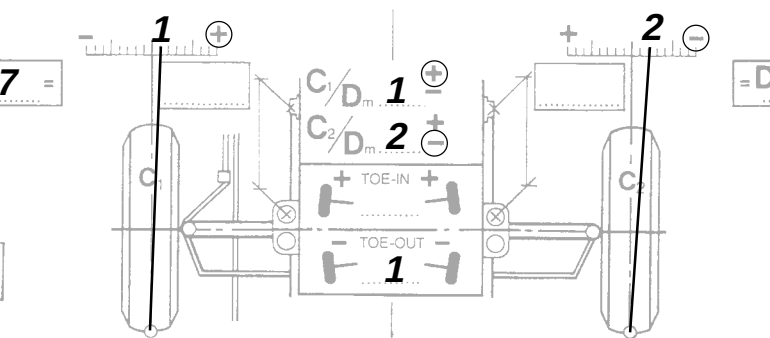
A₁
B₁
C₃ ± : D_m =

CAMBER
STURZ
CARROSSAGE
+

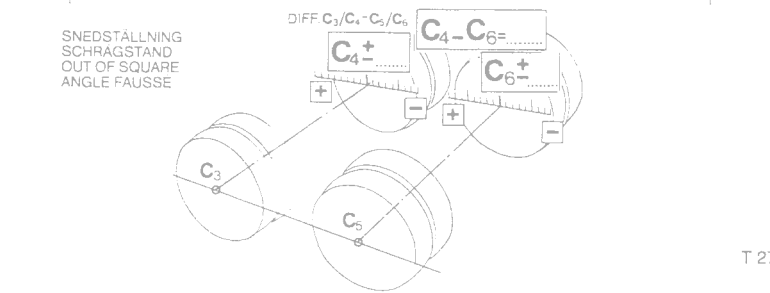
A₁
B₁
C₅ ± : D_m =

CAMBER
STURZ
CARROSSAGE
+

A₁ 0 100 200 300 **A₂ 300 200 100 0**



B₁ 0 100 200 300 **B₂ 300 200 100 0**



A₂ 140
B₂ 154
= D_m 7 : ± 14 C₂

CAMBER
STURZ
CARROSSAGE
+

KPI
Spreizung
INCLIN.
PIVOTS
+

+ CASTER
NACHLAUF
CHASSE

A₂
B₂
= D_m : ± C₄

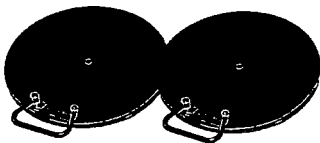
CAMBER
STURZ
CARROSSAGE
+

A₂
B₂
= D_m : ± C₆

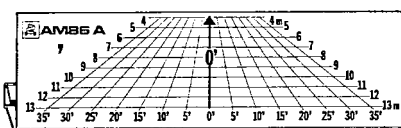
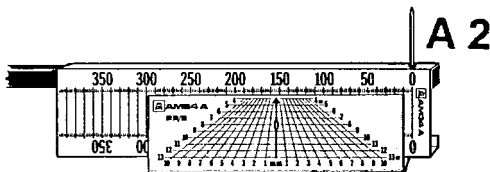
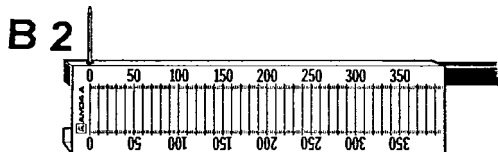
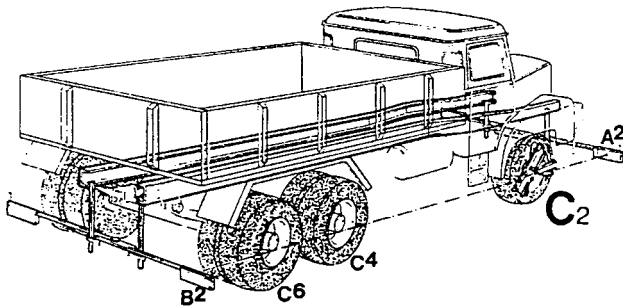
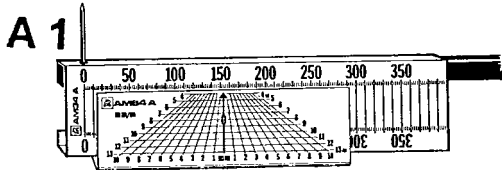
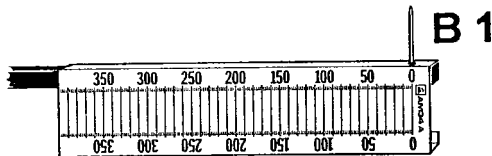
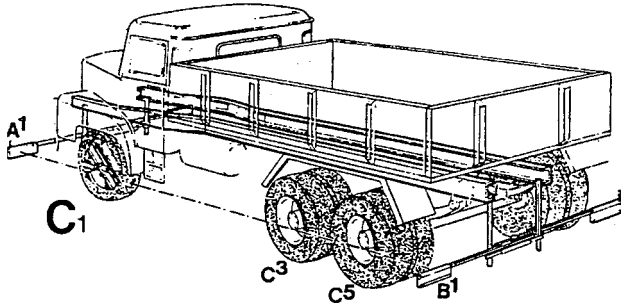
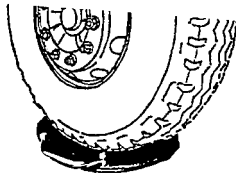
CAMBER
STURZ
CARROSSAGE
+



JOSAM laser AM Measuring system



Non-friction plates AM268



Measurement of front axles

Toe-in/toe-out with the sliding scale AM84 A

Hang the frame gauges, gauge extensions and measuring scales. Jack up the front axle.

Attach the wheel adapter and laser projector.

Compensate for run-out of the wheel with the adjustment knobs on the adapter.

Put first the left and then the right wheel in a straight ahead position.

The same value on measuring scales A and B. Lower the wheels onto the floor.

Put the sliding scales onto both the front measuring scales A1 and A2.

To make it easier to adjust toe-in/toe-out, it is an advantage to place both the wheels on non-friction plates AM268.

Measure the distance between the measuring scales A and B. Write down the result on the Test Report Sheet. **For example: D = 7m.**

Wheel C1 left side

Aim the laser dot at the measuring scale B1 left rear. Read the value. **For example: 151.**

Move the sliding scale on the measuring scale A1 front left and install the 0-line on 151.

Aim the laser dot on 7m on the horizontal distance line of the sliding scale. On the bottom row, read the rolling direction of the wheel +1 mm/m.

Write down 1 at C1/Dm in the middle part of the Test Report Sheet and circle a (+) sign. Draw in rolling direction of the wheel with a solid line.

Wheel C2 right side

Aim the laser dot at the measuring scale B2 right rear. Read the value. **For example: 154.**

Move the sliding scale on the measuring scale A2 right front and put the 0-line on 154.

Aim the laser dot at the 7m mark on the horizontal distance line of the sliding scale.

Read the rolling direction of wheel on the bottom row, -2 mm/m.

Write down 2 at C2/Dm in the middle part of the Test Report Sheet and circle a (-) sign. Draw in rolling direction of the wheel with a solid line.

If the laser dot lies inside the 0-line on the sliding scale, it is a (+) value.

If the laser dot lies outside the 0-line on the sliding scale, it is a (-) value.

Calculating toe-in/toe-out

$$C1/Dm = +1$$

$$C2/Dm = -2$$

$$\text{Result} = -1 \text{ (toe-out)}$$

Sliding scale AM86 A

The result on the bottom row is in minutes instead of mm/m.



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Reg. n° _____

Date _____

Miles/Km _____

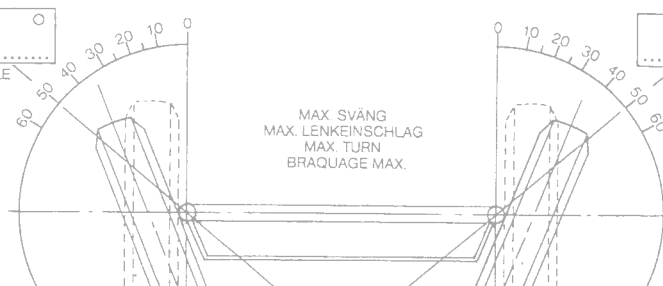
With sliding scale AM84 A
Toe-in/toe-out

KURVINKELDIFFERENS
SPURDIFFERENSWINKEL
TOE-OUT ON TURN
DIFFERENCE DE COURBE D'ANGLE

20° VÄNSTER
LINKS
LEFT
GAUCHE

HÖGER
RECHTS
RIGHT
DROIT

DIFF. _____



HÖGER
RECHTS
RIGHT
DROIT

20°

VÄNSTER
LINKS
LEFT
GAUCHE

DIFF. _____

A₁ 158

B₁ 151

C₁ 7 :D_m 7 =

CAMBER
STURZ
CARROSSAGE

KPI Spreizung
INCLIN.
PIVOTS

CASTER
NACHLAUF
CHASSE

A₁

B₁

C₃ ± :D_m =

CAMBER
STURZ
CARROSSAGE

A₁

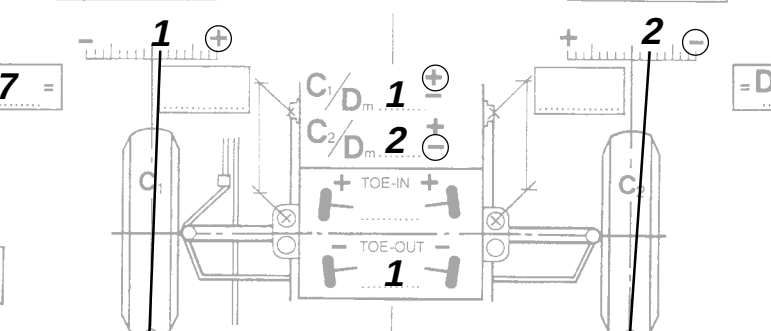
B₁

C₅ ± :D_m =

CAMBER
STURZ
CARROSSAGE

A₁ 0 100 200 300

A₂ 300 200 100 0



A₂ 140

B₂ 154

=D_m 7 :C₂ 14

CAMBER
STURZ
CARROSSAGE

KPI Spreizung
INCLIN.
PIVOTS

CASTER
NACHLAUF
CHASSE

A₂

B₂

=D_m :C₄

VÄNSTER
LINKS
LEFT
GAUCHE

HÖGER
RECHTS
RIGHT
DROIT

CAMBER
STURZ
CARROSSAGE

A₂

B₂

=D_m :C₆

VÄNSTER
LINKS
LEFT
GAUCHE

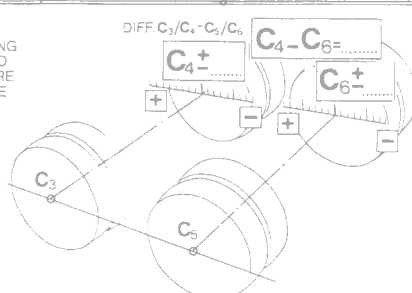
HÖGER
RECHTS
RIGHT
DROIT

CAMBER
STURZ
CARROSSAGE

B₁ 0 100 200 300

B₂ 300 200 100 0

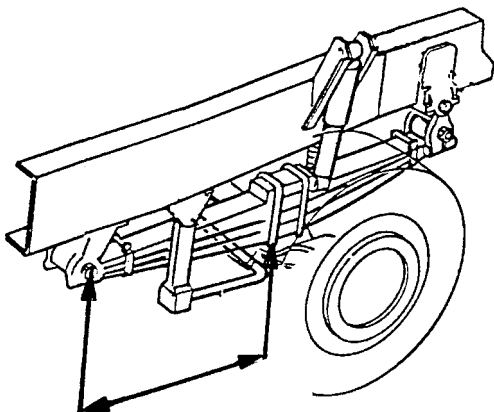
SNEDSTÄLLNING
SCHRAGSTAND
OUT OF SQUARE
ANGLE FAUSSE



A_{1.2}

B_{1.2}

=D_m



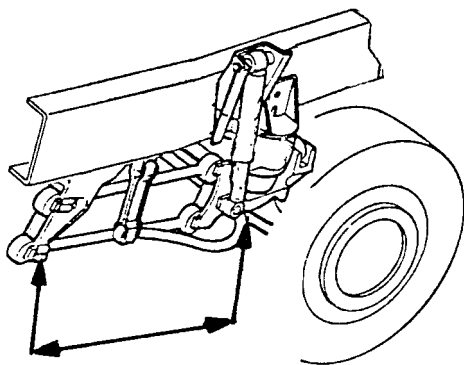
Measurement of out of square front axle

Check if a front axle with a steering spindle is out of square by using a tape measure.

Use the tape measure to measure the distance between the locating eye to a common reference point on the axle, i.e. from the spring-eye grease fitting to the u-bolt hole on the axle.

Write down the value on the Test Report Sheet.

Maximum allowed difference is 5 mm.



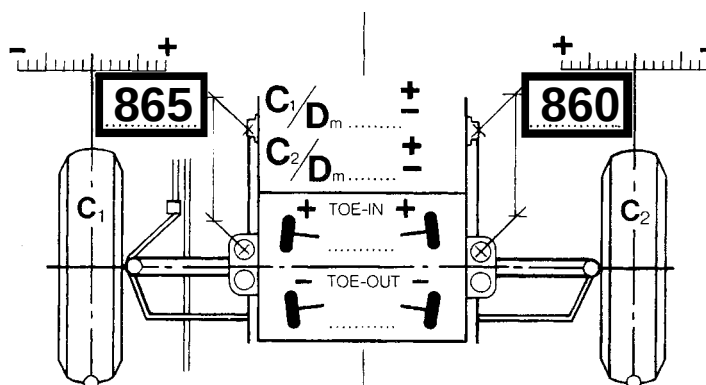
Example:

Left side	865 mm
Right side	860 mm
Difference	5 mm

A ₁	
B ₁	
C ₁	$\pm : D_m =$

CAMBER STURZ CARROSSAGE	KPI Spreizung
+	+
-	-

+	CASTER NACHLAUF CHASSE
-	



	A ₂
	B ₂
$= D_m$	$\pm : C_2$

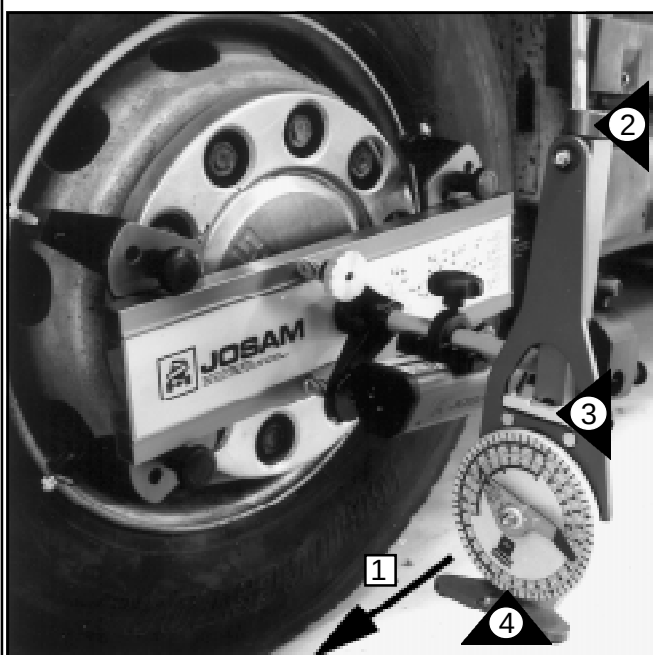
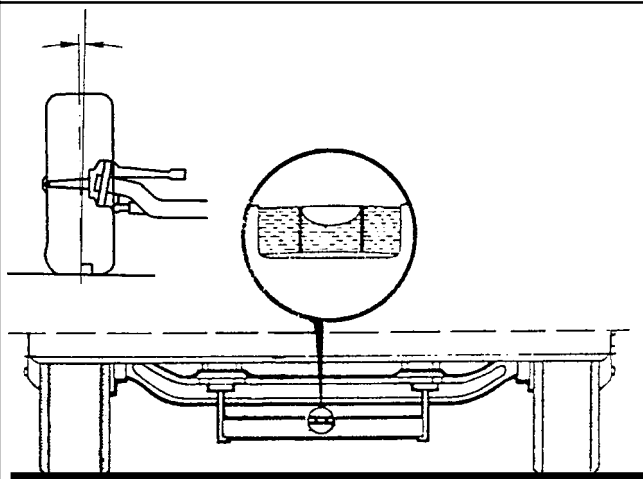
CAMBER STURZ CARROSSAGE	KPI Spreizung
+	+
-	-

+	CASTER NACHLAUF CHASSE
-	

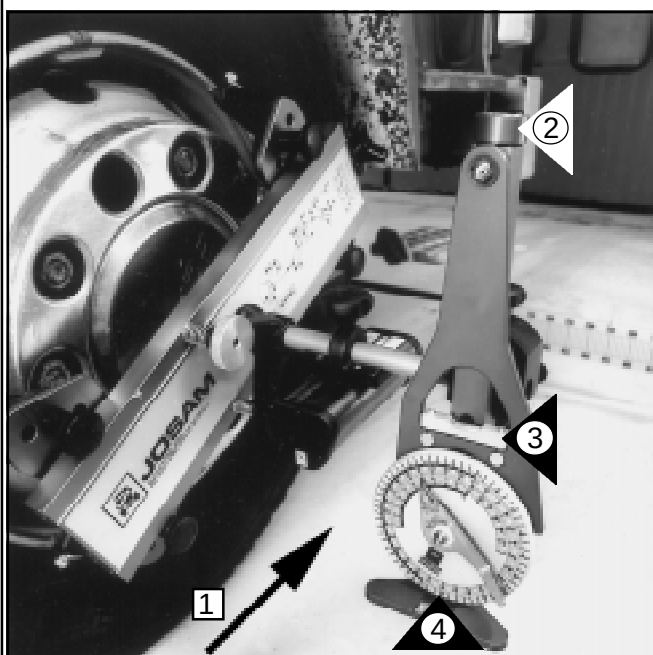


JOSAM laser AM Measuring system

Measurement of front axles



Left



Right

Measurement of camber

**When measuring camber, the axle must be level.
The wheel should be on the floor and "loaded".**

Hang the frame gauges, gauge extensions and measuring scales.

Jack up the front axle.

Attach the wheel adapter and laser projector.

Compensate for run-out of the wheel with the adjustment knobs on the adapter.

Attach the wheel alignment gauge AM300/301 onto the spindle of the wheel adapter through the hole for camber and caster measurements. Move the wheel alignment gauge toward the locking pin.

If the floor is not flat (level) or if the wheel sizes are different, check the position of the axle with a spirit level AM542.

The spirit level can be hung under the axle with a magnetic hanger, for example, AM98. Look for a place on the axle that is machined. Do not measure on the underside of the springs if there are different size spring spacers between the axle and the springs.

Adjust any unevenness by placing plates under one wheel until the axle is level.

If the floor is flat (level), check the measurement between the floor and the wheel hub, bead of rim or something similar.

- 1 Put the wheel in a straight ahead position.
Equal values on the front and rear measuring scales.
Lower the axle onto the floor.
- 2 Position the gauge vertically using the top bubble and lock into place with the lower knob.
- 3 Rotate the measuring scale so that the measuring bubble is level.
- 4 Read the value on the outer scale with the arrow on the foot plate.

Write down the value on the Test Report Sheet.
Note if it is a (+) or (-) value.

CAMBER STURZ CARROSSAGE	KPI Spreizung INCLIN. PIVOTS
+	+
-	-
CASTER NACHLAUF CHASSE	+
	-

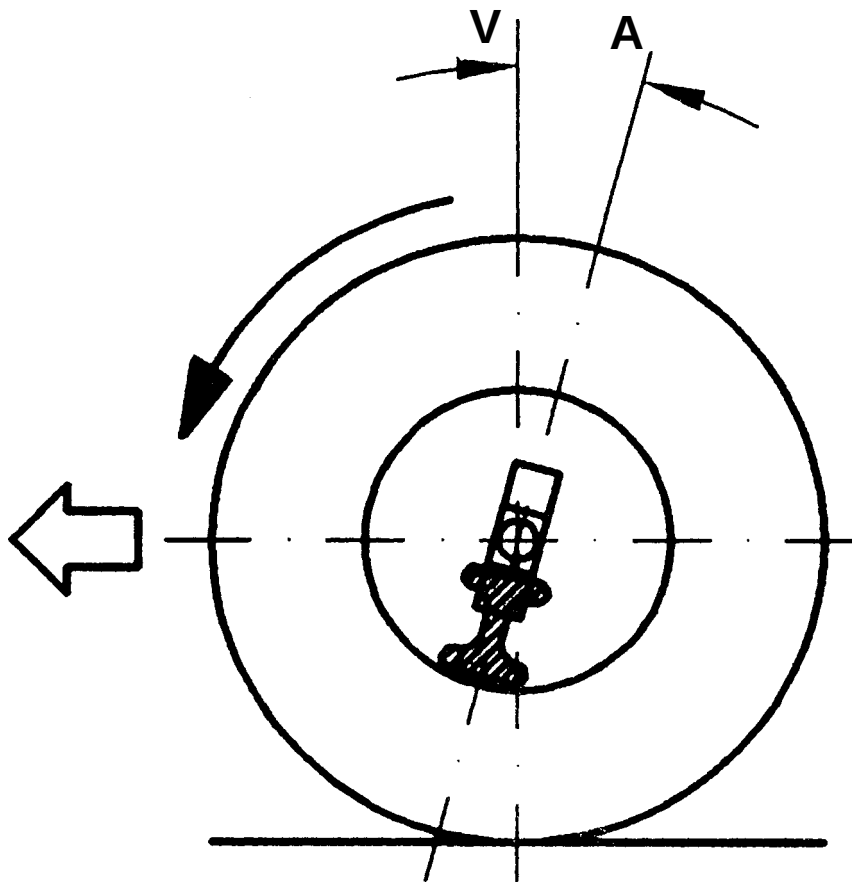
NOTE!

**When measuring camber the axle must be level.
The wheel should be on the floor and "loaded".**



Measurement of caster

V = Vertical line
A = Axle angle



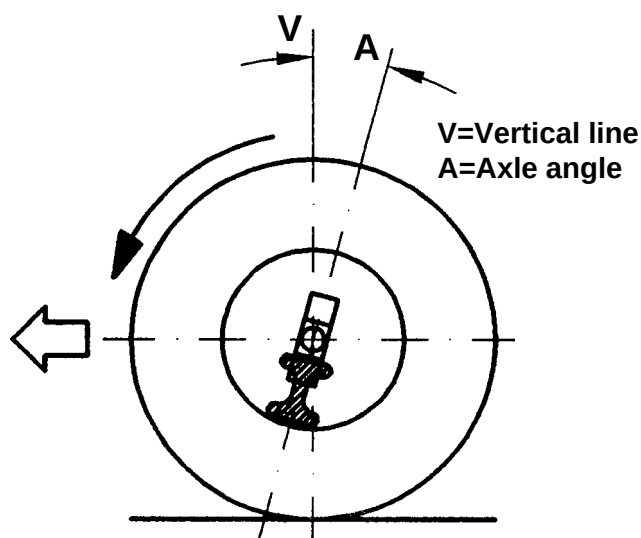
Caster

The inclination of the kingpin, backward or forward, as seen from the side.

When the kingpin's upper part inclines backward: it has positive (+) caster.

When the kingpin's upper part inclines forward: it has negative (-) caster.

Measure in degrees and minutes from the vertical line.



Measurement of caster

Use the wheel alignment gauge AM300/301 and turn angle gauge AM135.

Hang the frame gauges, gauge extensions and measuring scales. Jack up the axle so that the wheel rolls freely, approx. 10 mm above the floor. Attach the wheel adapter and laser projector. Compensate the run-out of the wheel with the adjustment knobs on the wheel adapter.

Lay the turn angle gauges AM135 on the floor for both the left and right sides, with a centered scale and in the middle of the wheels.

Put the left wheel on left-steered or the right wheel on right-steered vehicles in a straight ahead position with the laser projector. Equal values on the front and rear measuring scales.

Align the turn angle gauges on both sides with the laser dot on the 0-line of the measuring scale. Move the measuring scale along the profile and check at both ends that the laser dot hits the 0-line.

While measuring the caster with the help of the turn angle gauges AM135, the caster value will be insignificantly changed when the axle is lifted with the wheel approx. 10 mm above the floor.

NOTE!

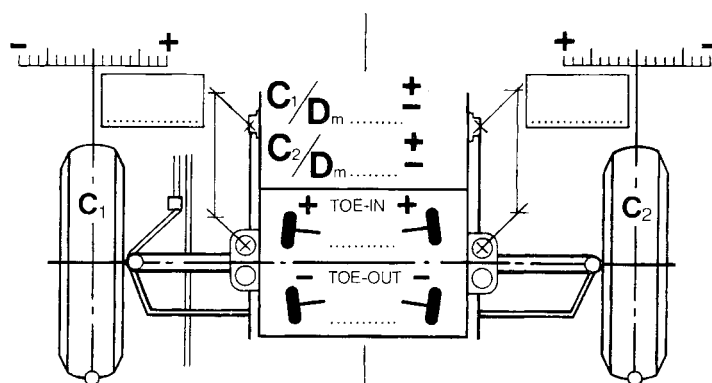
Bogie wheels should stay on the floor.



A ₁	
B ₁	
C ₁	$\pm$: D _m =

CAMBER STURZ CARROSSAGE	KPI Spreizung
$\pm$	$\pm$
$-$	$-$

$\pm$	CASTER NACHLAUF CHASSE
$-$	



	A ₂
	B ₂
$\pm$: D _m	C ₂

CAMBER STURZ CARROSSAGE	KPI Spreizung
$\pm$	$\pm$
$-$	$-$

$\pm$	CASTER NACHLAUF CHASSE
$-$	



JOSAM laser AM Measuring system

Measurement of front axles

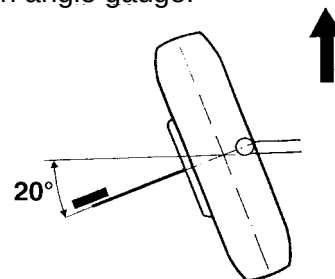
Measurement of caster

Use the wheel alignment gauge AM300/301 and turn angle gauge AM135.

Left side

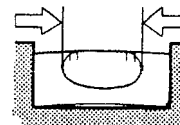
(1) Attach wheel alignment gauge AM300/301 on the spindle of the wheel adapter through the hole for camber and caster measurement. Move the wheel alignment gauge toward the locking pin.

(2) Turn the left wheel **20° outward** utilizing the turn angle gauge.



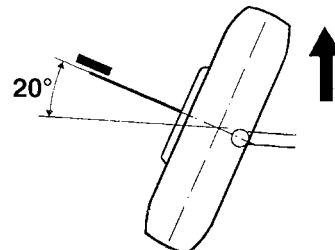
(3) Place the wheel alignment gauge vertically using the top bubble and lock with the bottom knob.

(4) Rotate the measuring scale so that the measuring bubble shows a horizontal position.



(5) Place the moveable pointer on **0** at the **inner** measuring scale.

(6) Rotate the left front wheel **20° inward** utilizing the turn angle gauge.

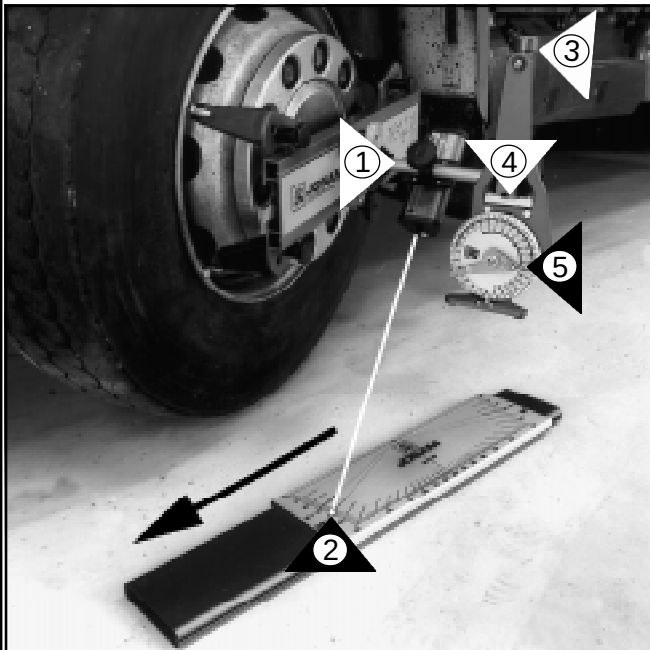


Place the wheel alignment gauge level using the top bubble and lock with the lower knob.

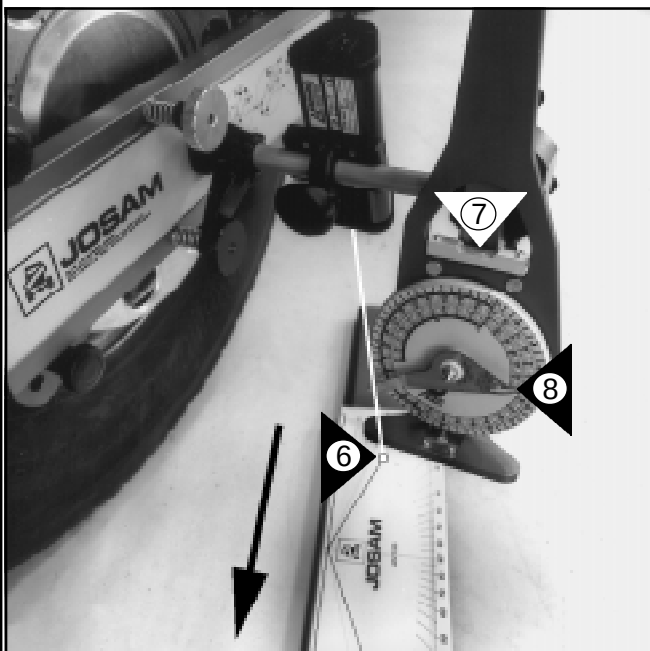
(7) Turn the measuring scale so that the measuring bubble shows a horizontal position.

(8) Read the caster value at the pointer on the **inner** measuring scale in **degrees ° and minutes ´**.

Write down the value on the Test Report Sheet.

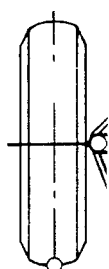


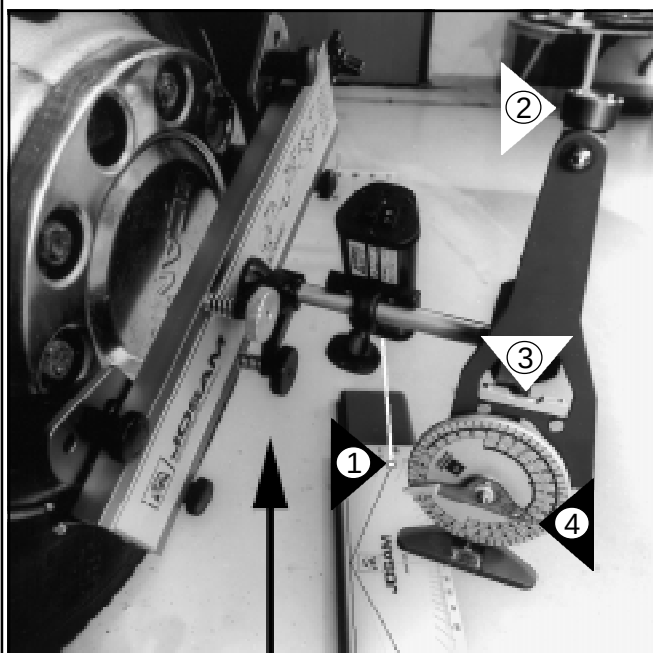
Left 20° outwards



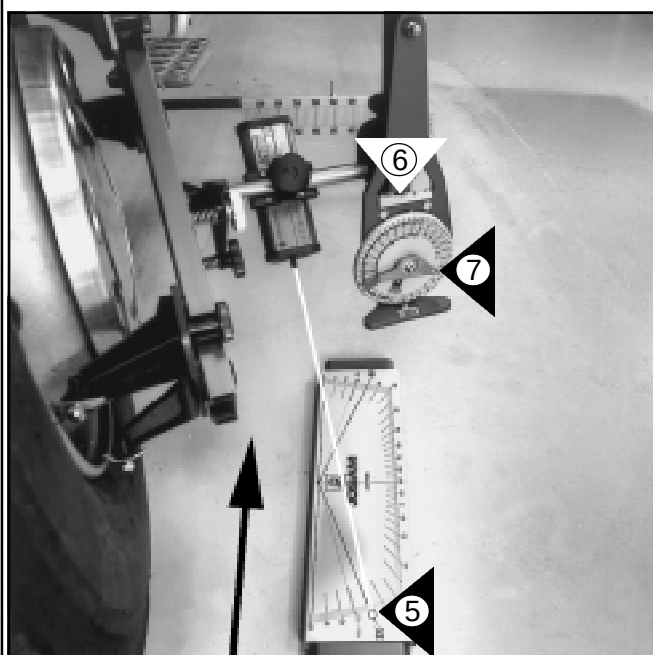
Left 20° inwards

CAMBER STURZ CARROSSAGE	KPI Spreizung
+	+
-	-
.....	
+	CASTER NACHLAUF CHASSE
-	
.....	





Right 20° outwards



Right 20° inwards



CAMBER STURZ CARROSSAGE	KPI Spreizung
+	+
-	-
.....	
CASTER NACHLAUF CHASSE	+
	-

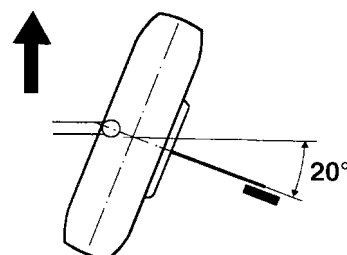
Measurement of caster

Use the wheel alignment gauge AM300/301 and turn angle gauge AM135.

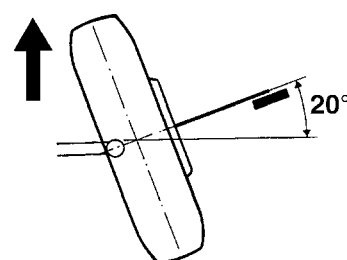
Right side

Attach wheel alignment gauge AM300/301 on the spindle of the wheel adapter.

- (1) Turn the left wheel **20° outward** utilizing the turn angle gauge.



- (2) Place the wheel alignment gauge vertically using the top bubble and lock with the bottom knob.
- (3) Rotate the measuring scale so that the measuring bubble shows a horizontal position.
- (4) Place the moveable pointer on **0** toward the **inner** measuring scale.
- (5) Rotate the right front wheel **20° inward** utilizing the turn angle gauge.



Place the wheel alignment gauge level using the top bubble and lock with the lower knob.

- (6) Turn the measuring scale so that the measuring bubble shows a horizontal position.
- (7) Read the caster value at the pointer on the **inner** measuring scale in **degrees ° and minutes '.**

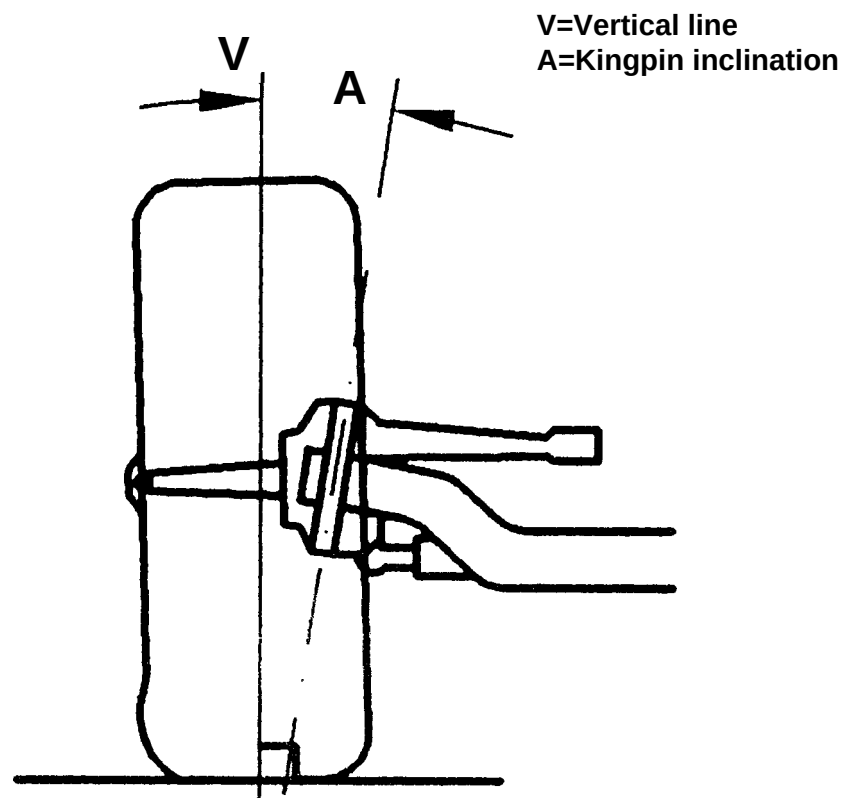
Write down the value on the Test Report Sheet.

Note!

When measuring the caster always begin with turning the wheel **20° OUTWARD** on both sides.



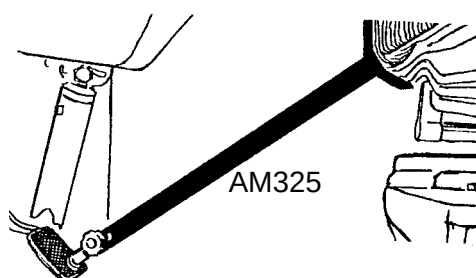
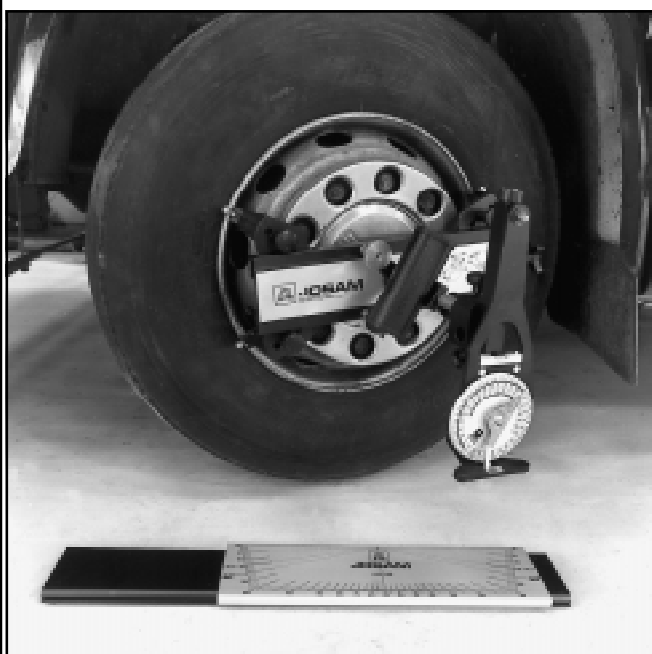
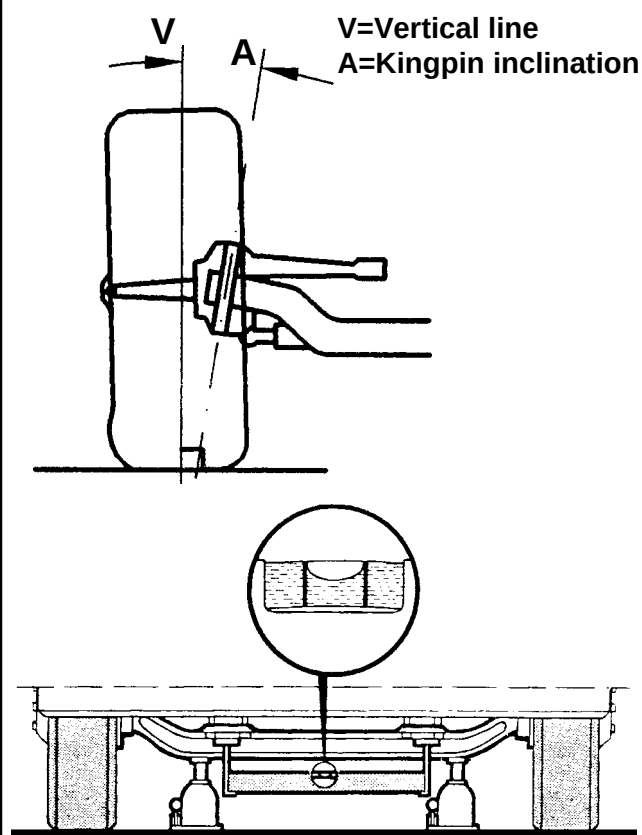
Measurement of KPI (kingpin inclination)



Kingpin inclination

The upper part of the kingpin inclines inward when seen from behind.

Measurement in degrees (°) and minutes (´) from the vertical line.



Measurement of KPI

Use the wheel alignment gauge AM300/301 and turn angle gauge AM135

Hang the frame gauges, gauge extensions and measuring scales.

Jack up the front axle.

Attach the wheel adapter and laser projector. Compensate for run-out of the wheel with the adjustment knobs on the adapter.

The axle should be jacked up when measuring KPI with the help of turn angle gauges AM135.

When measuring KPI the axle must be level. The wheel should also have a brake engaged with brake rod AM325.

Check the position of the axle with a spirit level. The spirit level can be hung under the axle with a magnetic hanger, for example, AM98. Look for a place on the axle that is machined. Do not measure on the underside of the springs if there are different size spring spacers between the axle and the springs.

It is easy to make the axle level with the help of two jacks.

When the axle is level, place the turn angle gauges AM135 on the floor for both the left and right sides, with the scales centered and lined up with the middle of the wheels.

Place the left and then the right wheel in a straight ahead position with the laser projector. Equal values on the front and rear measuring scales.

Position the turn angle gauges on both sides with the laser dot on the 0-line of the measuring scale.

Move the measuring scale along the profile and check at both ends that the laser dot hits the 0-line.

NOTE!

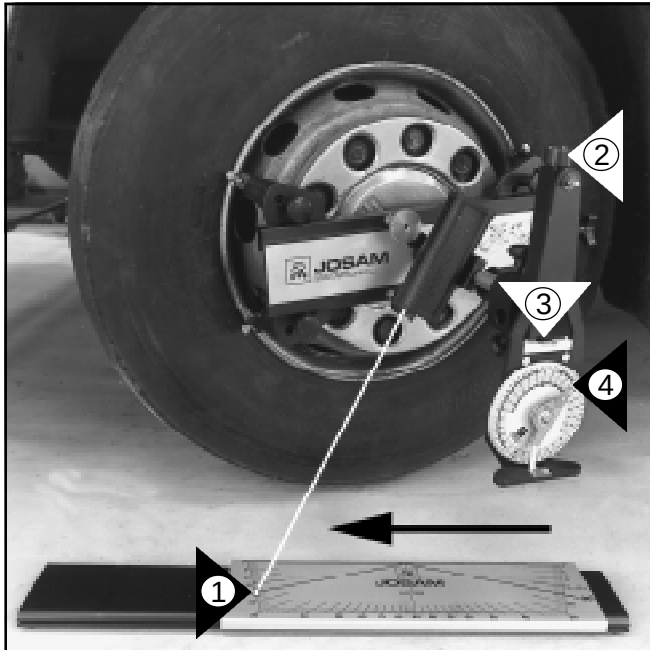
When taking KPI measurements, the axle must be level and the wheel must have the brake engaged.

Brake rod AM325 placed between the brake pedal and the seat may be conveniently used.

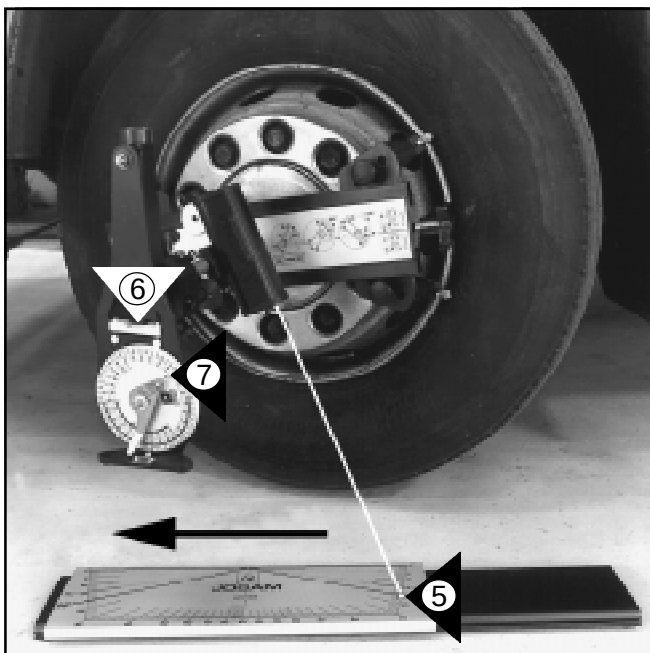


JOSAM laser AM Measuring system

Measurement of front axles

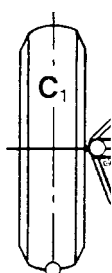


Left 20° outwards



Left 20° inwards

CAMBER STURZ CARROSSAGE	KPI Spreizung
+	+
-	-
.....	
+	CASTER NACHLAUF CHASSE
-	
.....	



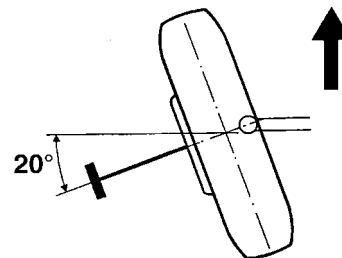
Measurement of KPI

Use the wheel alignment gauge AM300/301 and turn angle gauge AM135.

Left side

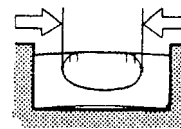
Attach the wheel alignment gauge AM300/301 onto the spindle of the wheel adapter through the hole for KPI measurements. Slide the wheel alignment gauge in until it stops on the spindle.

- (1) Turn the left front wheel **20° outward** (to the left) utilizing the turn angle gauge.



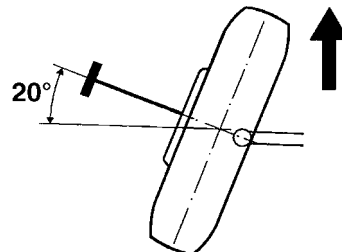
- (2) Put the wheel alignment gauge in a vertical position using the upper level and lock it into place with the upper knob.

- (3) Turn the measuring scale so that the bubble indicates a horizontal position.



- (4) Position the moveable pointer on **0** toward the inner scale.

- (5) Turn the left front wheel **20° inward** (to the right) utilizing the turn angle gauge.



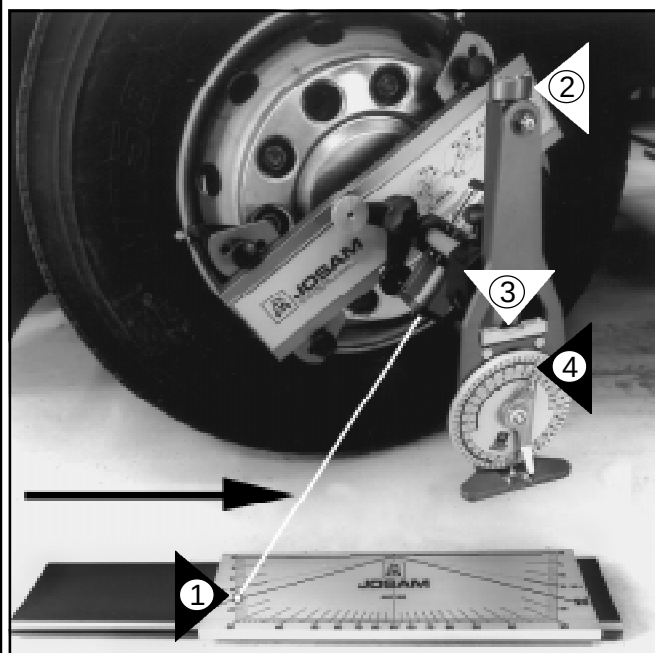
NOTE!

The measuring gauge should not be placed in a horizontal position.

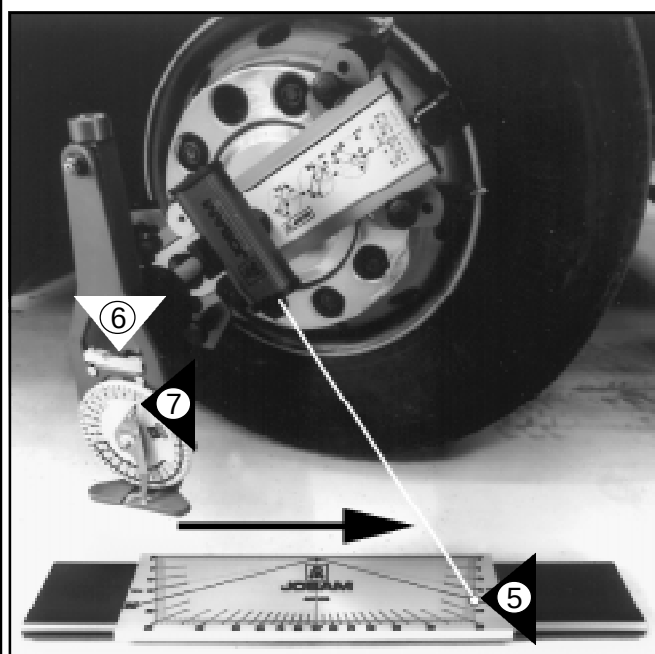
- (6) Turn the measuring scale so that the bubble shows a horizontal position.

- (7) Read the KPI value at the pointer on the **inner** measuring scale in **degrees °** and **minutes '.**

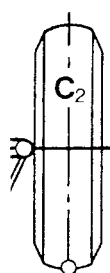
Write down the value on the Test Report Sheet.



Right 20° inwards



Right 20° outwards



CAMBER STURZ CARROSSAGE	KPI Spreizung
+	+
-	-
CASTER NACHLAUF CHASSE	+
	-

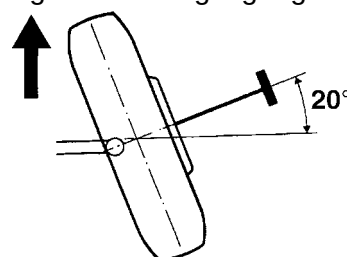
Measurement of KPI

Use the wheel alignment gauge AM300/301 and turn angle gauge AM135.

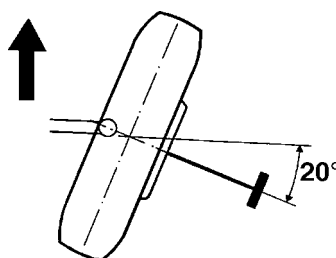
Right side

Attach the wheel alignment gauge AM300/301 onto the axle using a wheel adapter.

- (1) Turn the left front wheel **20° inward** (to the left) utilizing the turn angle gauge.



- (2) Put the wheel alignment gauge in a vertical position using the top bubble and lock it into place with the upper knob.
- (3) Turn the measuring scale so that the main bubble indicates a horizontal position.
- (4) Position the moveable pointer on **0** toward the **inner** scale.
- (5) Turn the left front wheel **20° outward** (to the right) utilizing the turn angle gauge.



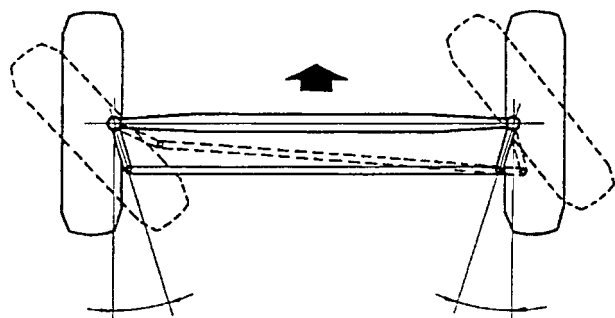
NOTE!

The measuring gauge should not be placed in a horizontal position.

- (6) Turn the measuring scale so that the main bubble shows a horizontal position.
- (7) Read the KPI value at the pointer on the **inner** measuring scale in **degrees °** and **minutes ´**. Write down the value on the Test Report Sheet.

NOTE!

When measuring KPI, always begin by turning the wheel 20° to the **LEFT** on both sides. Left side **OUTWARD** and right side **INWARD**.



Measuring toe-out on turn with the turn angle gauge AM135.

To check if the angle of the steering arm is correct or if the arm has been damaged, the toe-out on turn must be measured.

Some manufacturers determine the toe-out on turn by turning the inner wheel to 20° and then taking a reading on the outer wheel. Some determine the toe-out on turn at the maximum turn.

Hang the frame gauges, gauge extensions and measuring scales.

Jack up the front axle.

Attach the wheel adapter and laser projector. Compensate for run-out of the wheel with the adjustment knobs on the adapter.

Place the turn angle gauges AM135 on the floor for both the left and right sides, with a centered scale and in the middle of the wheels.

Put the left wheel on left-steered or the right wheel on right-steered vehicles in a straight ahead position with the laser projector. Equal values on the front and rear measuring scales.

Position the turn angle gauges on both sides with the laser dot on the 0-line of the measuring scale.

Turn the inner wheel 20° outward.

Then check the angle of the outer wheel on the opposite side. Position the laser dot on the measuring scale. Aim the laser dot on the scale and read the value.

Example: left wheel: 20° (outward)

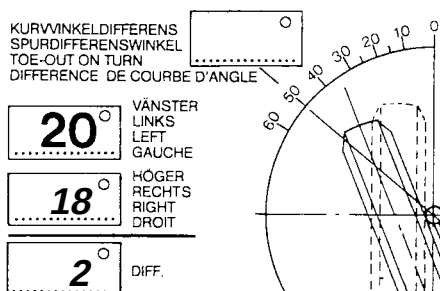
right wheel: 18° (inward)

Difference: 2°

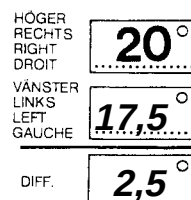
Write down the value on the Test Report Sheet and calculate the toe-out on turn.

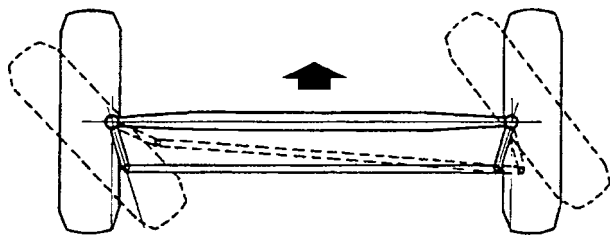
Repeat the procedure on the opposite side.

The maximum allowed difference is 0.5° (30') between the left and right sides.



MAX. SVÄNG
MAX. LENKEINSCHLAG
MAX. TURN
BRAQUAGE MAX.





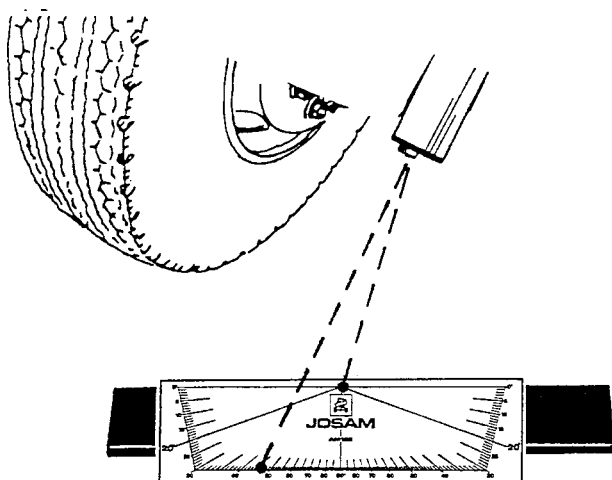
Measurement of maximum turn with the turn angle scales AM135.

Hang the frame gauges, gauge extensions and measuring scales.

Jack up the front axle.

Attach the wheel adapter and laser projector. Compensate for run-out of the wheel with the adjustment knobs on the adapter.

Place the turn angle gauges AM135 on the floor for both the left and right sides, with a centered scale and in the middle of the wheels.



Put the left wheel on left-steered or the right wheel on right-steered vehicles in a straight ahead position with the laser projector. Equal values on the front and rear measuring scales.

Position the turn angle gauges on both sides with the laser dot on the 0-line of the measuring scale.

Turn the wheel out as far as possible.

Aim the laser dot on the 0-line of the measuring scale. Slide the scale until the laser dot hits the intersection point on the 0-line.

Aim the laser dot on the outer (degrees) scale and read the value.

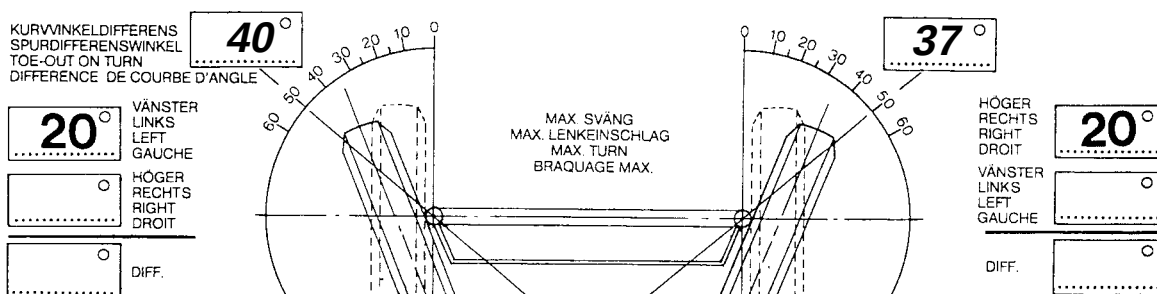
Write down the value on the Test Report Sheet.

Compare the measurement results with each other and with the manufacturer's specifications.

Example: Left side 40°
Right side 37°

If the maximum turn is too great, it must be corrected.

Caution must be taken that the steering linkage angle doesn't become 180° causing the steering to lock!



This instruction is issued by:



JOSAM

Maskingatan 5 • Box 419 • S-701 48 ÖREBRO, SWEDEN

Phone: +46 19-30 40 00 • Fax: +46 19-32 03 16

E-mail: info@josam.se • Internet: www.josam.se

Representative: